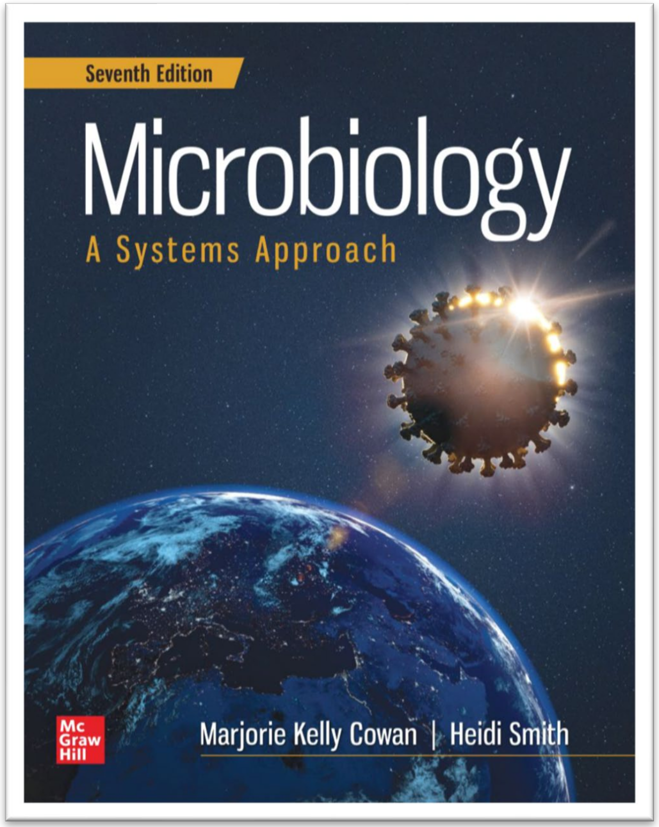
Chapter 21:
Infectious Diseases Manifesting in the Cardiovascular and Lymphatic Systems

Cilicia Nascimento

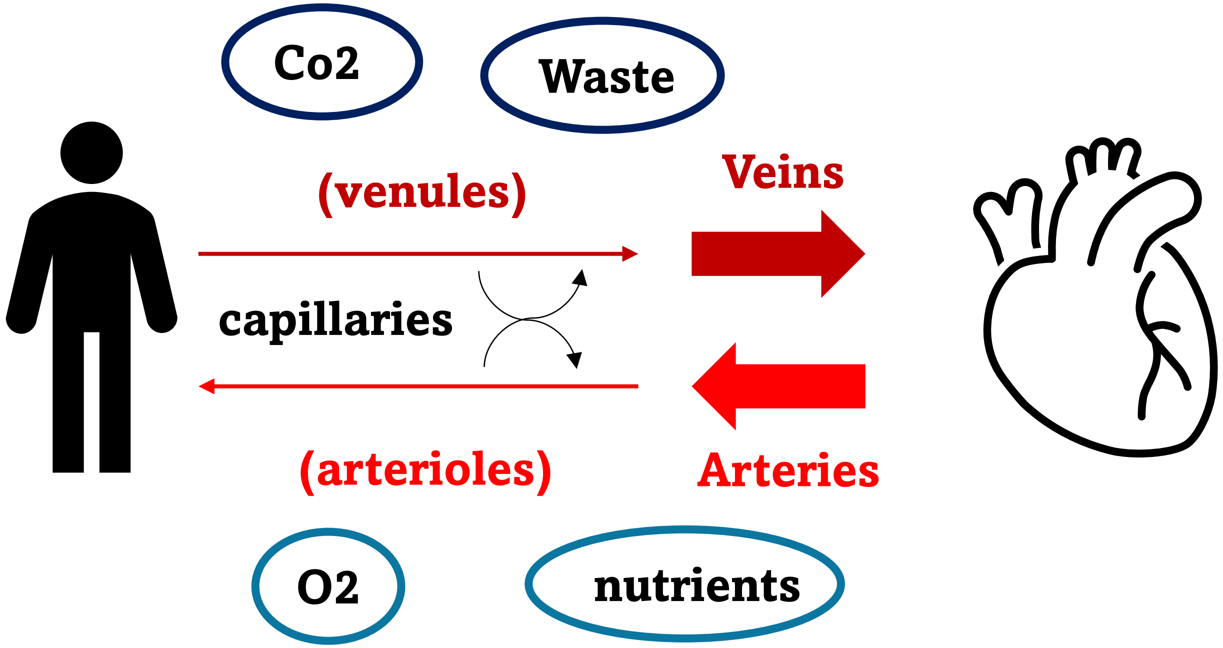


Microbiology
A Systems Approach
Seventh Edition
Marjorie Kelly Cowan, Heidi Smith

Learning outcomes (21.1)

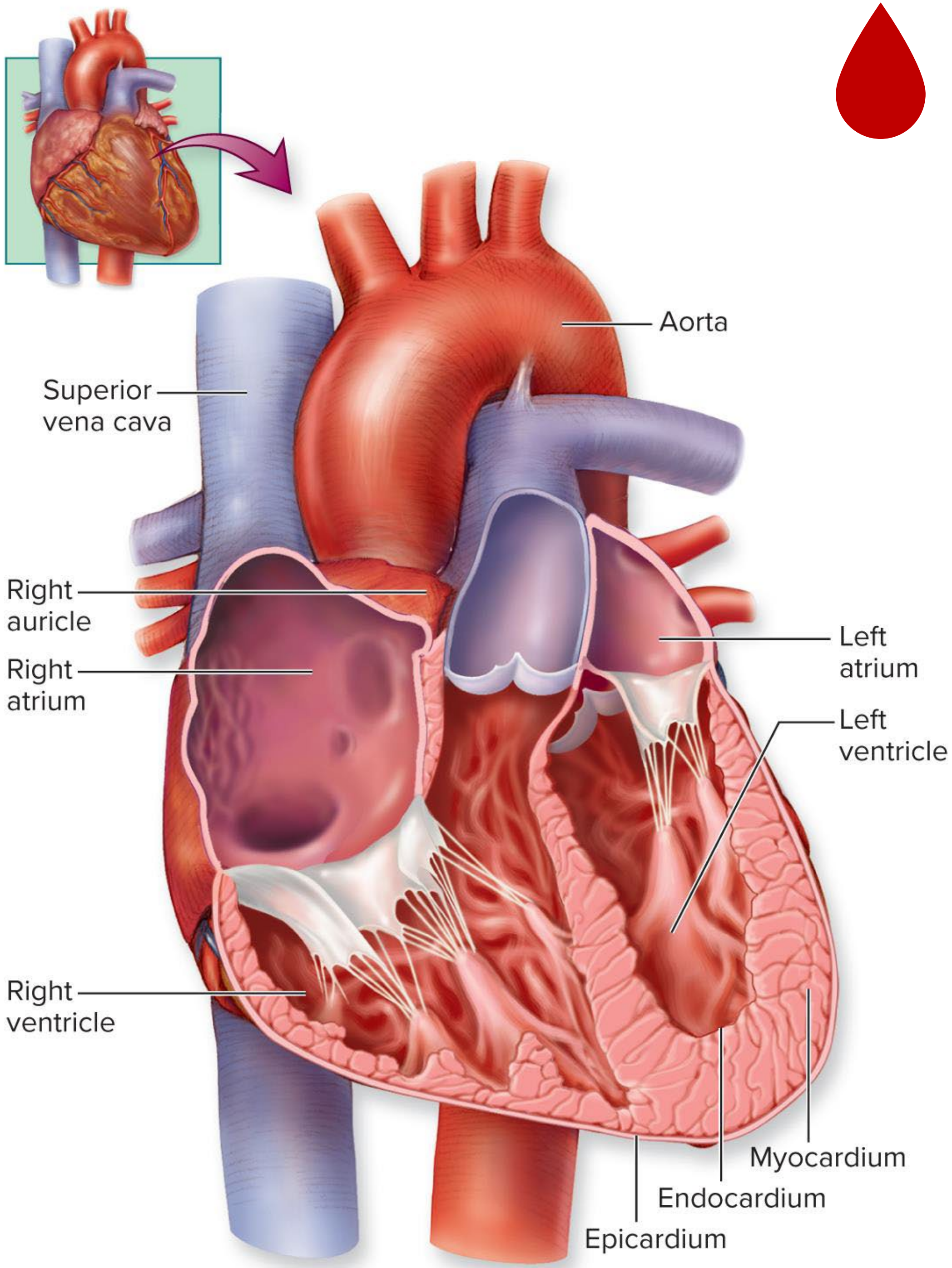
- Describe the important **anatomical features** of the cardiovascular system.
- List the **natural defenses** present in the cardiovascular and lymphatic systems.
- Discuss the current state of knowledge of the **normal biota** of the cardiovascular and lymphatic systems.

The cardiovascular system

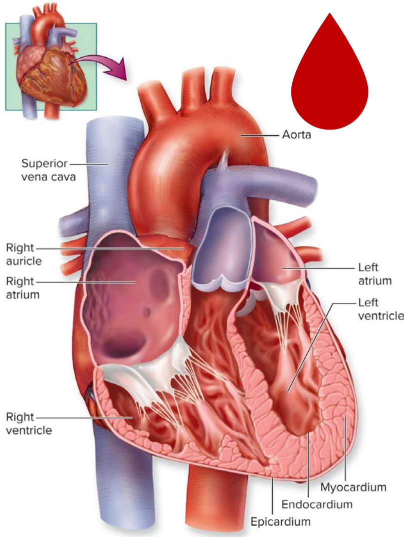


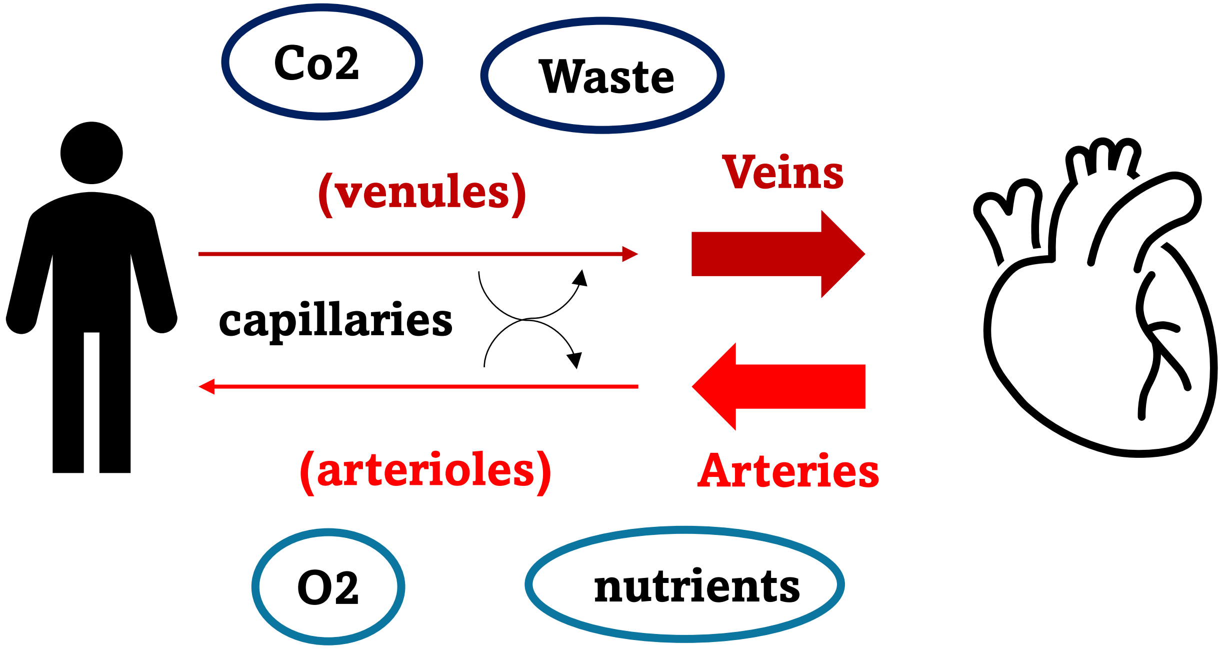
White Blood Cells (WBCs)
- **Macrophages**
- **Neutrophils**
- **Basophils**
- **Eosinophils**

Innate and Adaptive immune response



The cardiovascular system





Vir**emia**: Viruses that cause meningitis
Fung**emia**: Fungi in the blood
Bacter**emia**: *Presence* of bacteria in the blood
Septic**emia**: bacteria flourishing and growing in the bloodstream.

White Blood Cells (WBCs)

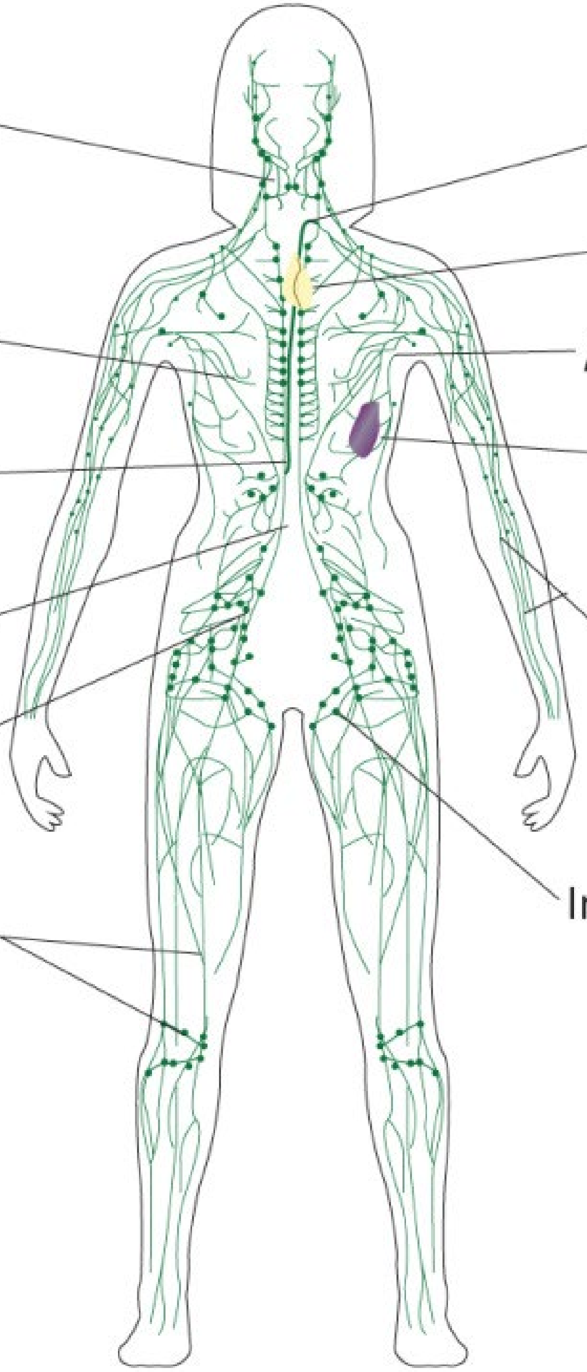
- **Macrophages**
- **Neutrophils**
- **Basophils**
- **Eosinophils**

Innate and Adaptive immune response

The lymphatic system

- Source of immune cells
  - Lymphocytes
    B cells
    T cells
    NK cells
- Lymph vessels
- Lymph nodes
- Tonsils
- Spleen
- Thymus
- Bone Marrow

Adaptive



Null M, Arbor TC, Agarwal M. Anatomy, Lymphatic System. [Updated 2023 Mar 6]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK513247/

Normal biota

- Microorganism present, but no normal microbiome
- Low-level microbial "infections" may contribute to diseases for which no infectious cause has been found

The Healthy Human Blood Microbiome: Fact or Fiction?

*Diego J. Castillo[1], Riaan F. Rifkin[1,2], Don A. Cowan[1] and Marnie Potgieter[1]**

[1] Department of Biochemistry, Genetics and Microbiology, Centre for Microbial Ecology and Genomics, University of Pretoria, Pretoria, South Africa, [2] Human Origins and Palaeo Environmental Research Group, Department of Anthropology and Geography, Oxford Brookes University, Oxford, United Kingdom

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part 2

Blood pathogens

Learning outcomes (21.2)

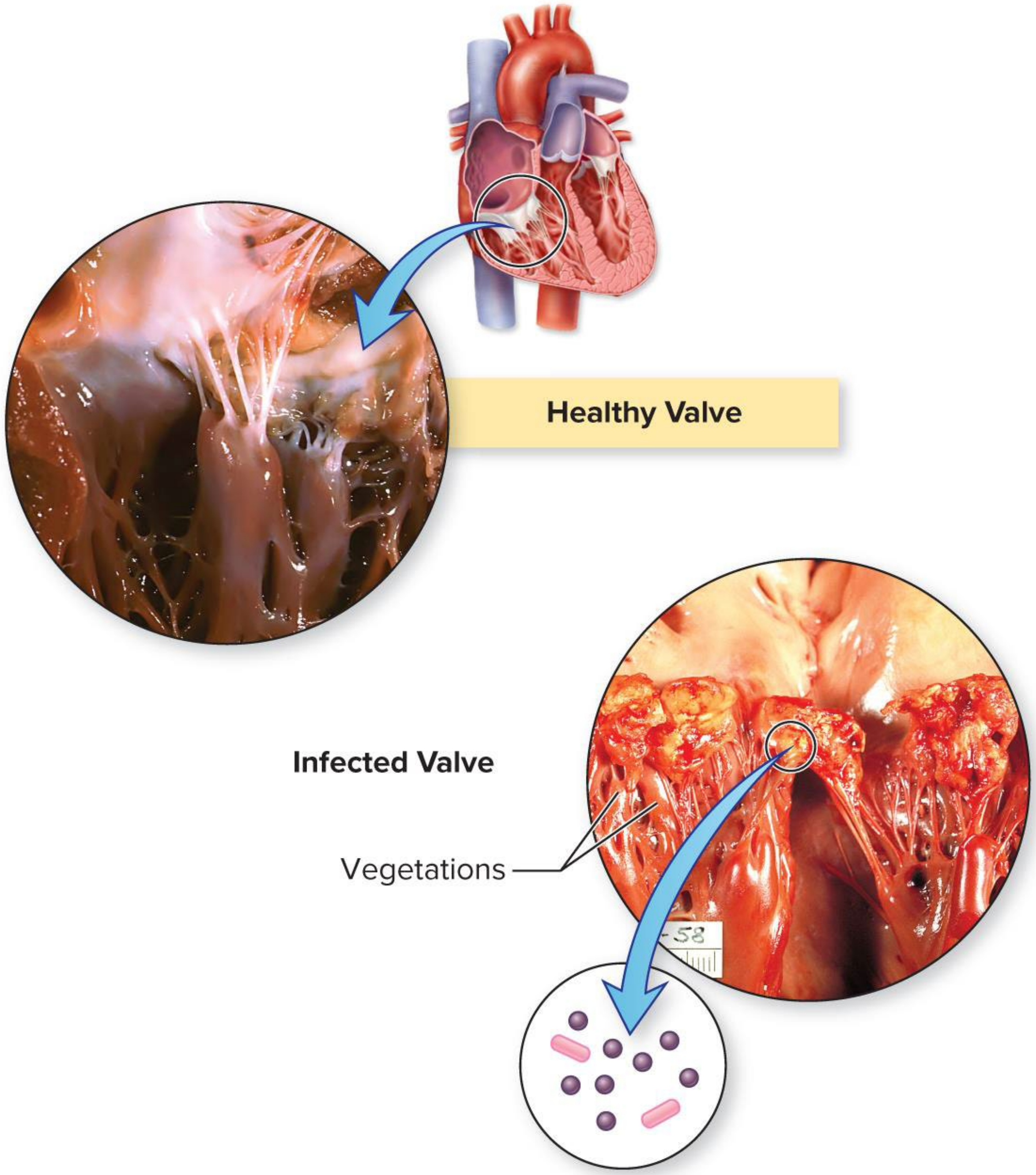
- List the possible causative agents for each of the following infectious cardiovascular conditions: **COVID-19, endocarditis, plague, tularemia, Lyme disease, infectious mononucleosis, anthrax, Chagas disease, and malaria**.
- Explain why **COVID-19** is placed in the **cardiovascular chapter**.
- Discuss what series of events may lead to **sepsis** and how it should be **prevented** and **treated**.
- Discuss the difference between **hemorrhagic** and **nonhemorrhagic fever** diseases.

Learning outcomes (21.2)

- List the possible **causative agents** and modes of transmission for **hemorrhagic fever** diseases.
- List the possible **causative agents** and modes of transmission for **nonhemorrhagic fever** diseases.
- Identify the five or six most relevant facts about **malaria**.
- Describe important **events** in the course of an **HIV infection** in the absence of treatment.
- Discuss the **epidemiology** of **HIV** infection in the United States and in the developing world.

Endocarditis

- Inflammation of the inner lining of the heart (endocardium)
- Damage to heart valves or prosthetic heart valves predisposes patients to endocarditis
- Acute
- <u>Subacute</u>
  - Fever, anemia, abnormal heartbeat, symptoms of heart attack, shortness of breath, and chills
  - Abdominal or side pain, Janeway lesions, and Osler's nodes
  - <u>Enlarged spleen, clubbed fingers, and toes</u>



Infective Endocarditis Disease Table

Disease	Acute Endocarditis	Subacute Endocarditis
Causative Organism(s)	*Staphylococcus aureus, Streptococcus pyogenes, S. pneumoniae, Enterococcus, Pseudomonas aeruginosa*, others	Alpha-hemolytic streptococci, others
Most Common Modes of Transmission	Parenteral	Endogenous transfer of normal biota to bloodstream
Culture/Diagnosis	Blood culture	Blood culture
Prevention	Aseptic surgery, injections	Prophylactic antibiotics before invasive procedures
Treatment	Vancomycin; surgery	Broad-spectrum antibiotics; surgery may be necessary
Distinctive Features	Acute onset, high fatality rate	Slower onset
Epidemiological Features	Greatly increased incidence due to heroin epidemic	N/A

Sepsis x Endotoxic Shock

1. Occurs when organisms are actively multiplying in the blood
2. Result of gram-negative bacteria multiplying in the bloodstream and releasing endotoxin
3. Leads to a drastic loss of blood pressure
4. Can be caused by many different bacteria and a few fungi
5. Increased breathing rate, respiratory alkalosis, and low blood pressure resulting in loss of fluid from the vasculature
6. Fever, altered mental state, shaking chills, and gastrointestinal symptoms

Sepsis x Endotoxic Shock

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6. Fever, altered mental state, shaking chills, and gastrointestinal symptoms (sepsis)

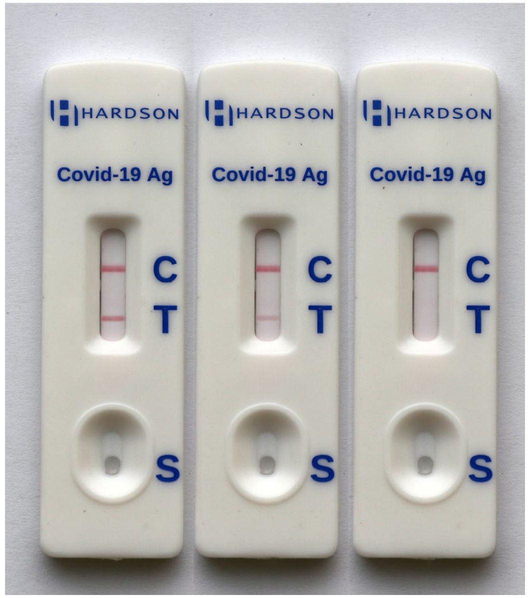
Sepsis Disease Table

Causative Organism(s)	Bacteria or fungi
Common Modes of Transmission	Parenteral, endogenous transfer
Virulence Factors	Cell wall or membrane components
Culture/Diagnosis	Blood culture, deep sequencing
Prevention	N/A
Treatment	Broad-spectrum antibiotic until identification and susceptibilities tested. *C. auris* is in **Urgent Threat** category in CDC Antibiotic Resistance Report
Epidemiological Features	In United States: 1.7 million cases and 270,000 deaths per year

Let’s talk about COVID-19

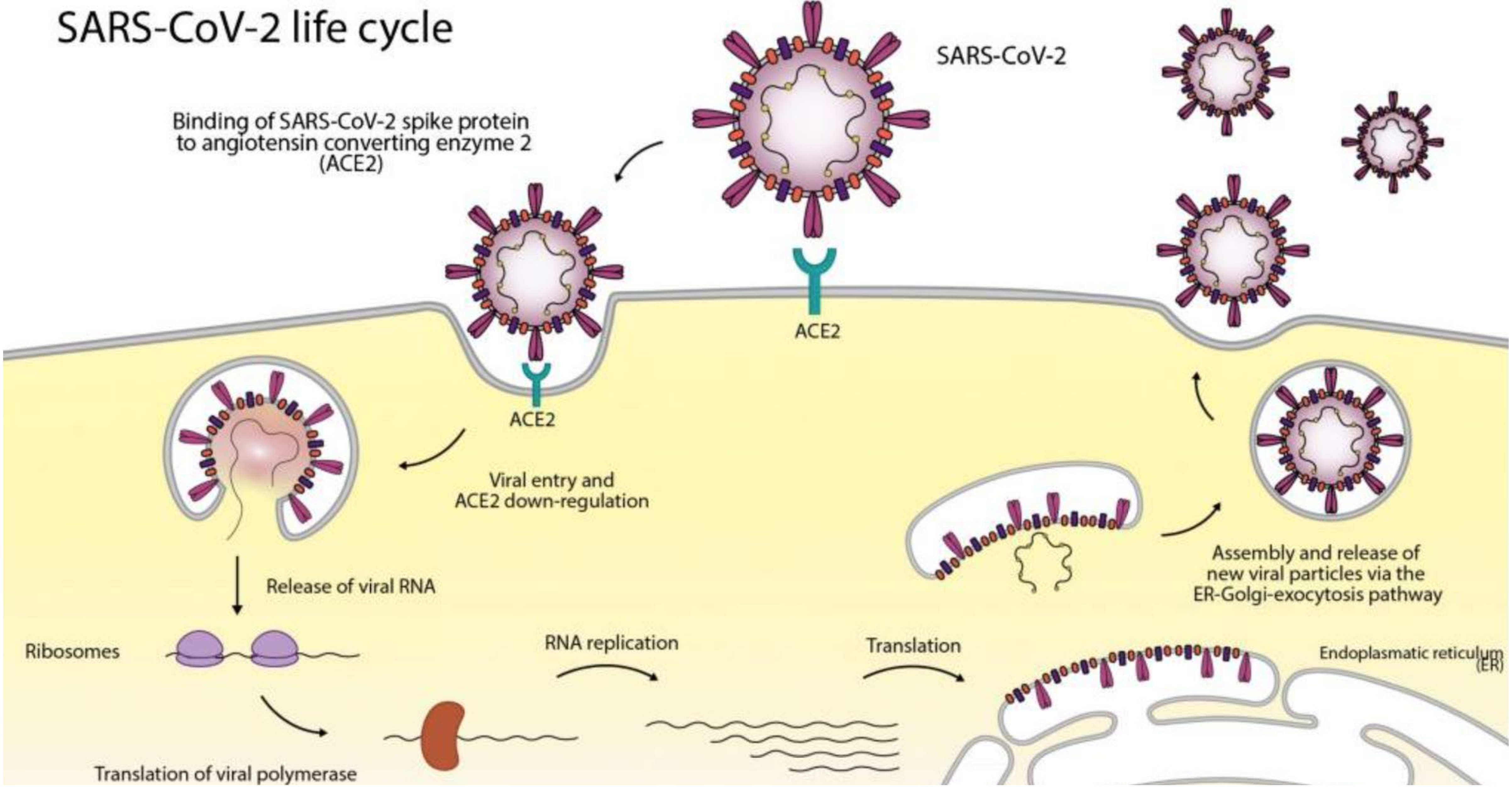
Let’s talk about COVID-19

- Respiratory virus
- Mild to severe symptoms
- Heart damage
- Blood clots
- One third of infections is asymptomatic
- Swab + PCR
- Vaccines available
- Paxlovid



- **Fever or chills**
- **Cough**
- **Sore throat**
- **Congestion or runny nose**
- **New loss of taste or smell**
- **Fatigue**
- **Muscle or body aches**
- **Headache**
- **Nausea or vomiting**
- **Diarrhea**
- **Trouble breathing**
- **Persistent pain or pressure in the chest**
- **New confusion**
- **Inability to wake or stay awake**
- **Depending on skin tone, lips, nail beds and skin may appear pale, gray, or blue.**

SARS-CoV-2 life cycle



Beyerstedt S, Casaro EB, Rangel ÉB. COVID-19: angiotensin-converting enzyme 2 (ACE2) expression and tissue susceptibility to SARS-CoV-2 infection. Eur J Clin Microbiol Infect Dis. 2021 May;40(5):905-919. doi: 10.1007/s10096-020-04138-6. Epub 2021 Jan 3. PMID: 33389262; PMCID: PMC7778857.

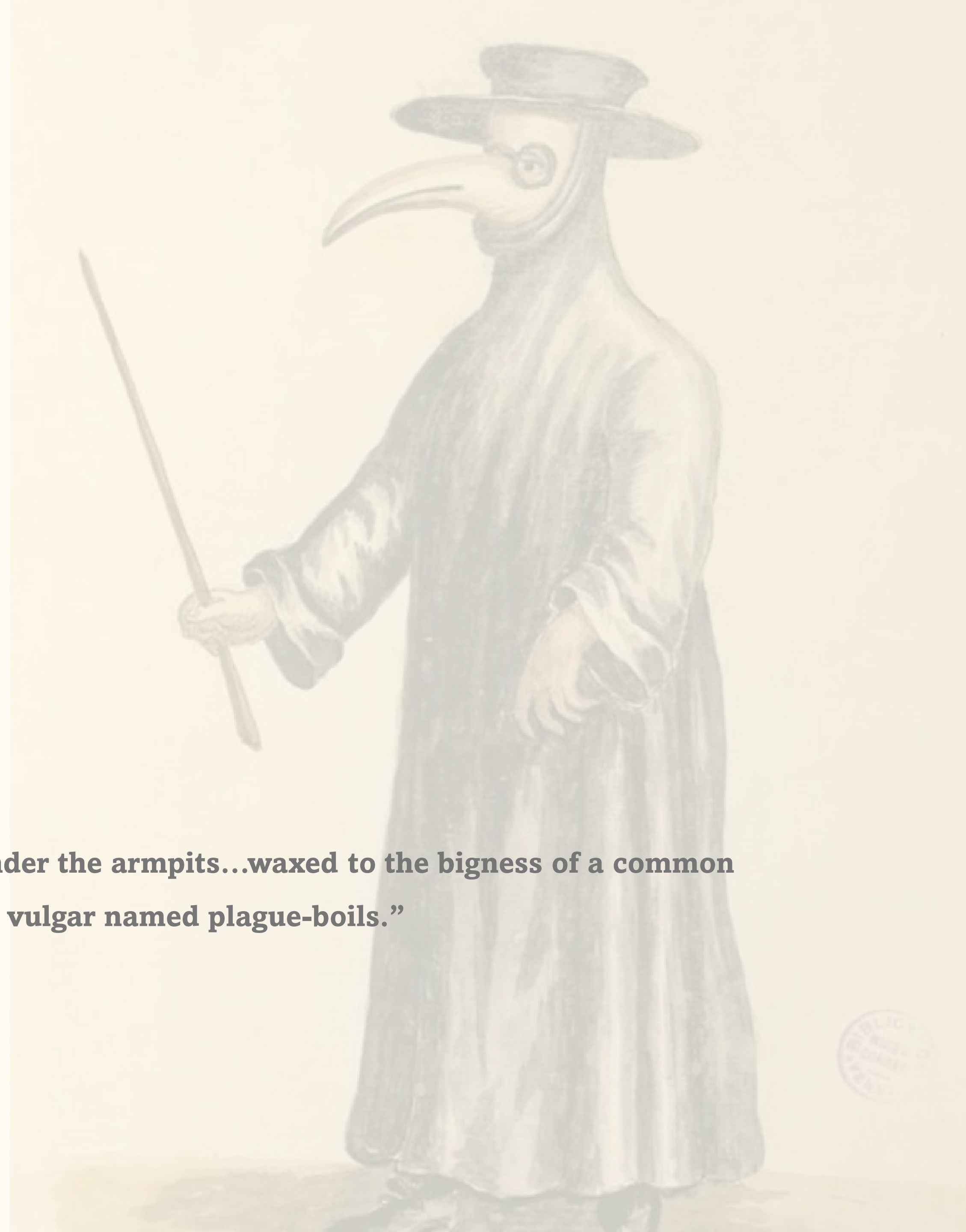
Covid-19 Table

Causative Organism(s)	SARS-CoV-2
Most Common Modes of Transmission	Droplet, airborne
Virulence Factors	Attachment to ACE-2; induction of autoimmunity
Culture/Diagnosis	RT-PCR, Ab and Ag tests
Prevention	Vaccine, mitigation efforts
Treatment	Antivirals such as paxlovid™
Epidemiological Features	Spreading constantly worldwide

The plague

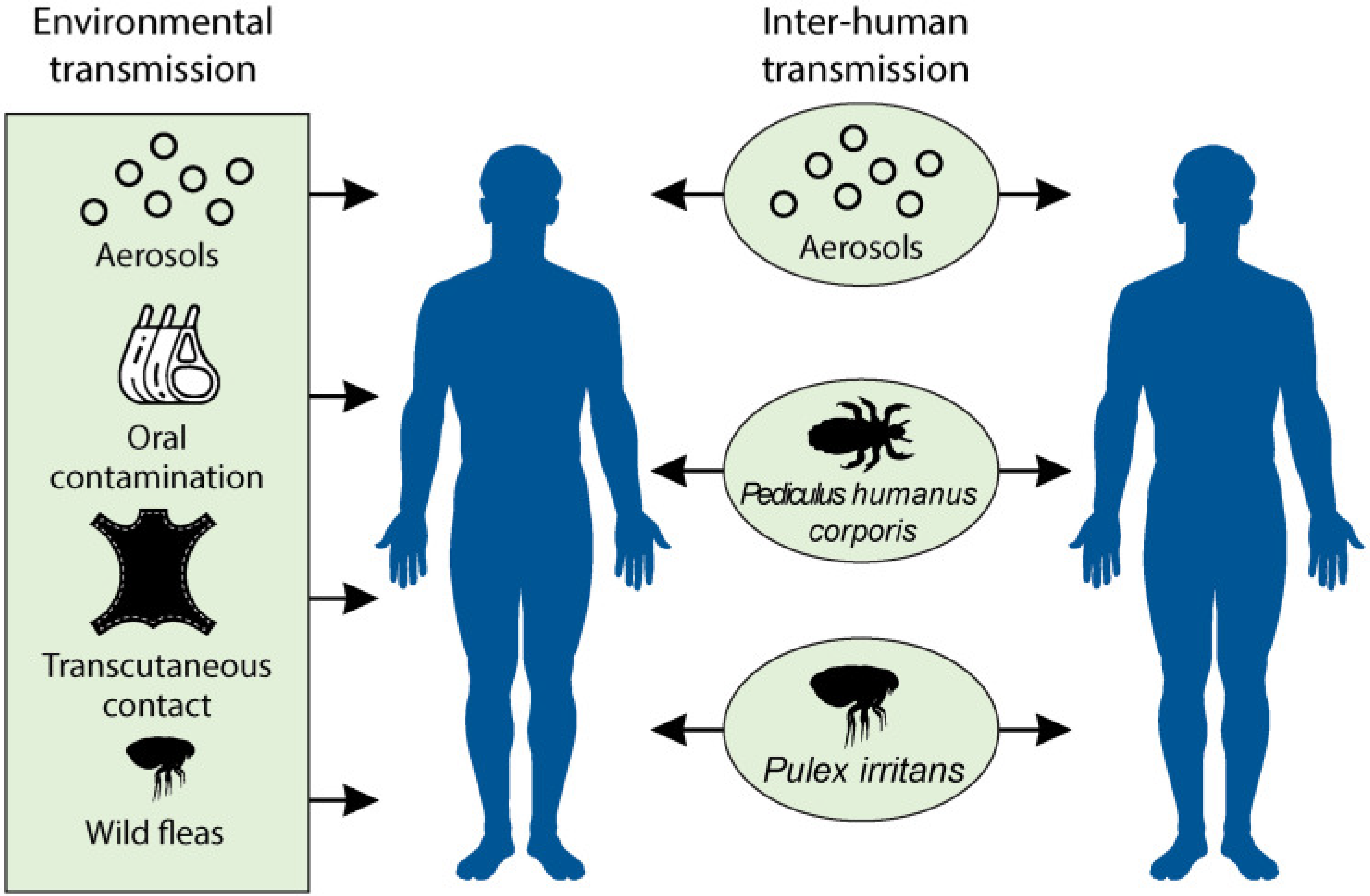
“at the beginning of the malady, certain swellings, either on the groin or under the armpits…waxed to the bigness of a common apple, others to the size of an egg, some more and some less, and these the vulgar named plague-boils.”

Giovanni Boccaccio



Bubonic plague

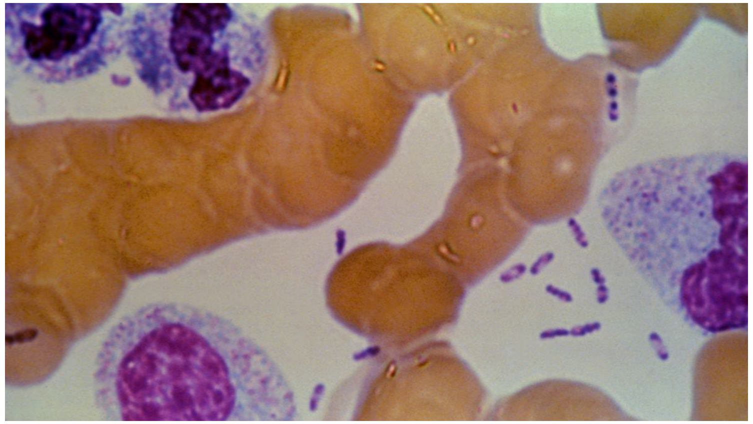
- Airborne bacteria



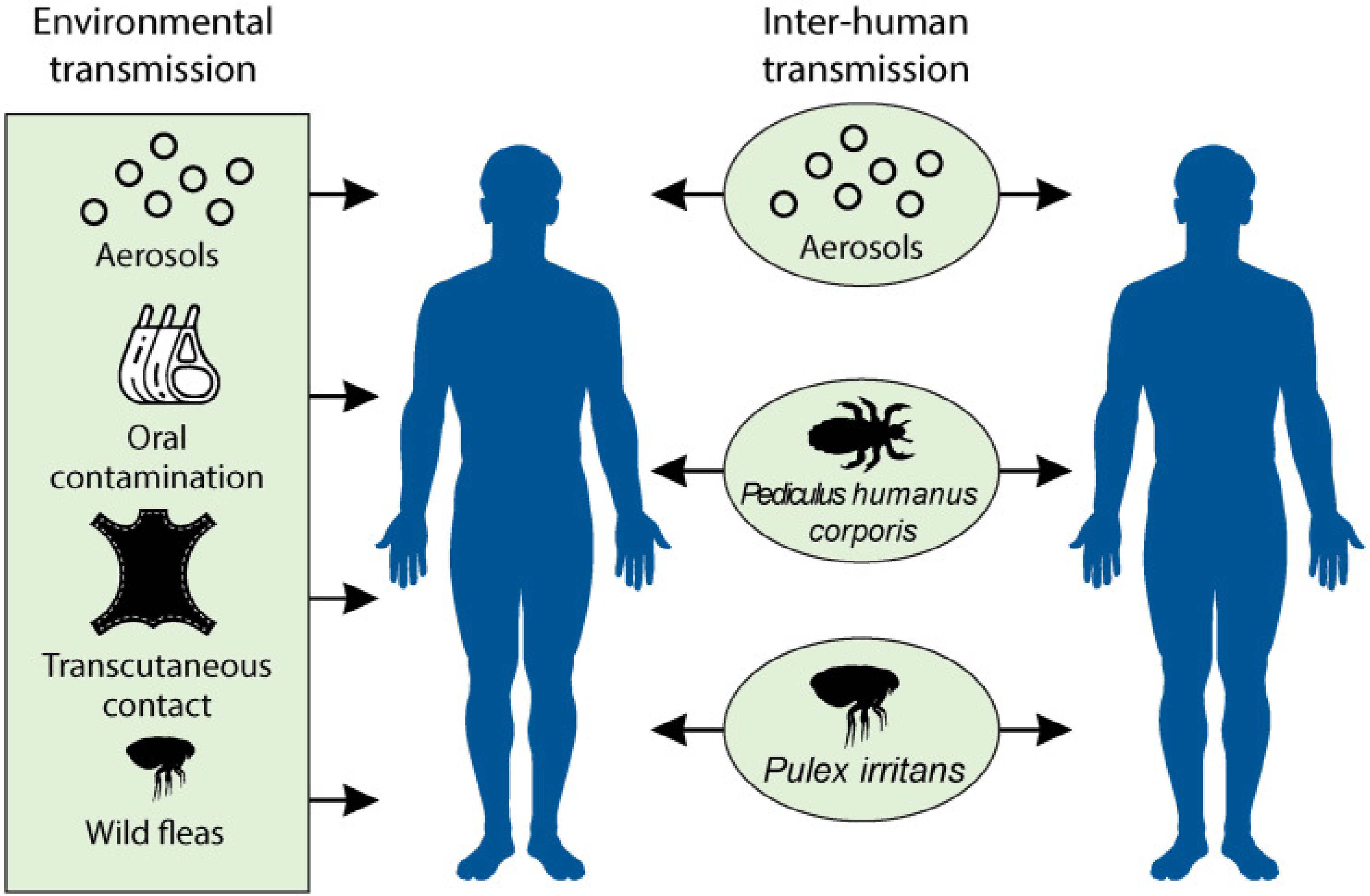
Barbieri R, Signoli M, Chevé D, Costedoat C, Tzortzis S, Aboudharam G, Raoult D, Drancourt M. Yersinia pestis: the Natural History of Plague. Clin Microbiol Rev. 2020 Dec 9;34(1):e00044-19. doi: 10.1128/CMR.00044-19. PMID: 33298527; PMCID: PMC7920731.

Bubonic plague

- Airborne bacteria
- *Yersinia pestis*



Gram-negative coccobacillus
3-50 required to initiate infection



Bubonic plague

- Airborne bacteria

- *Yersinia pestis*

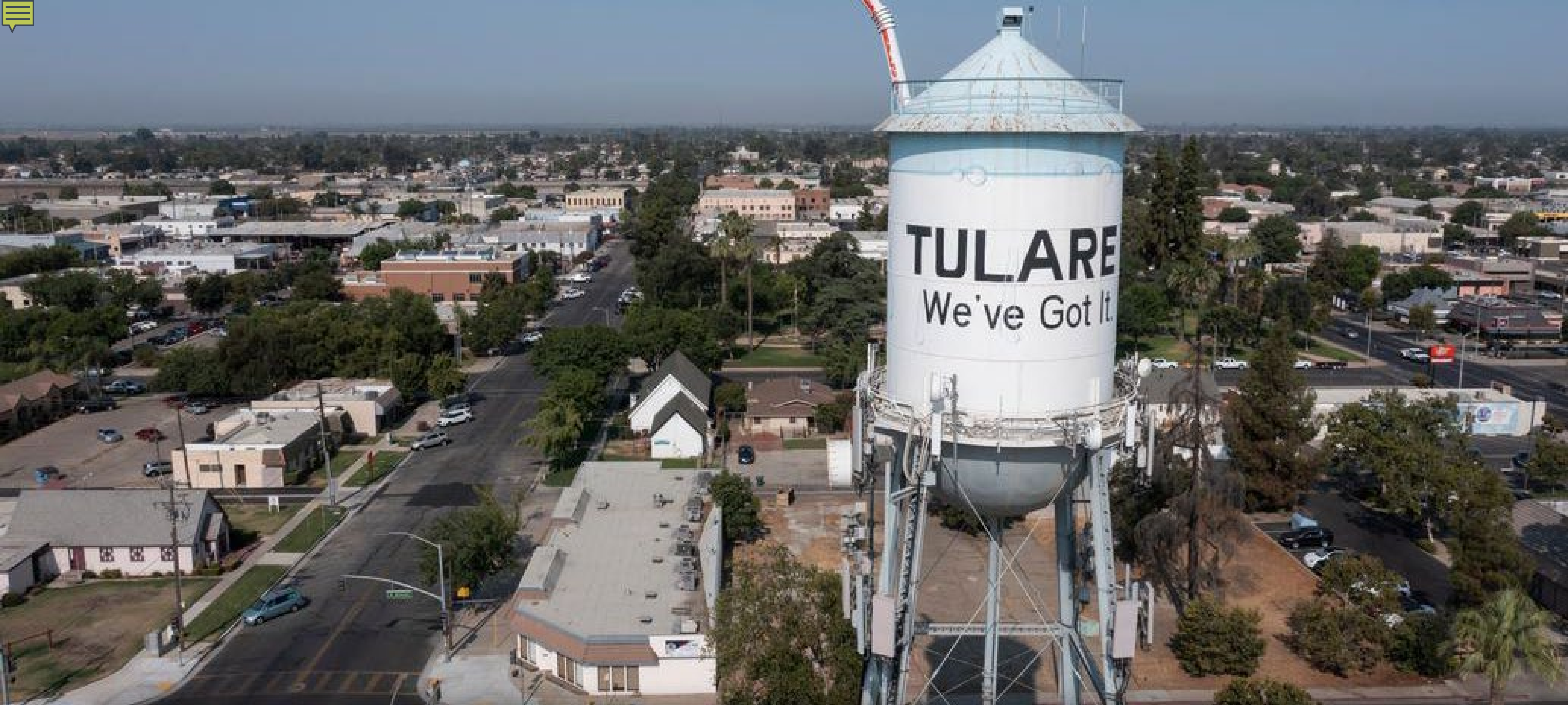
- Mortality
  - 30-50% with treatment (streptomycin/ciprofloxacin)
  - No treatment: 100%

- Symptoms

Bubonic plague Septicemic plague Pneumonic plague

Plague Table

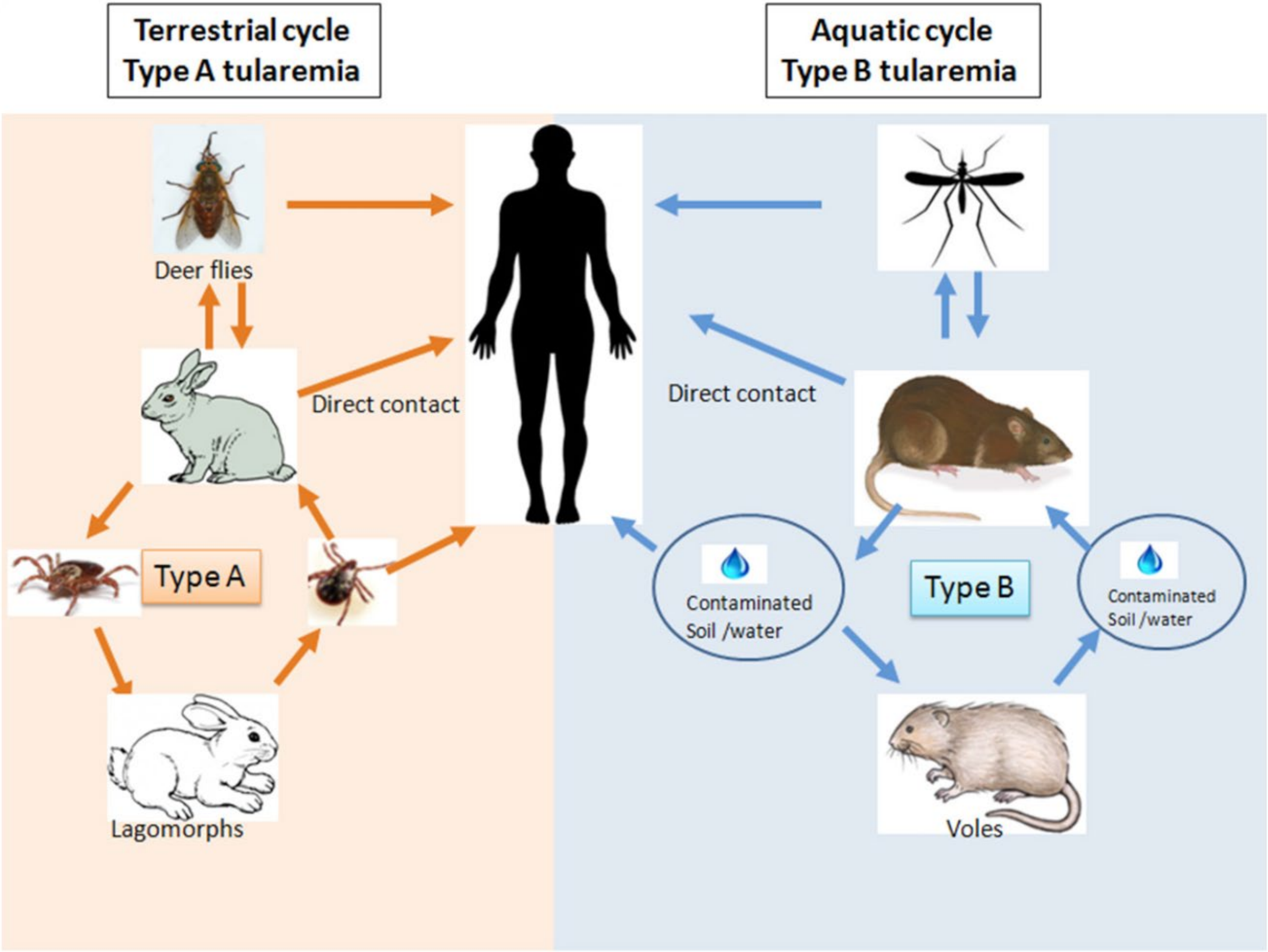
Causative Organism(s)	*Yersinia pestis*
Most Common Modes of Transmission	Biological vector (flea) also droplet contact (pneumonic) and direct contact with body fluids
Virulence Factors	Capsule, plasminogen activator
Culture/Diagnosis	Rapid genomic methods
Prevention	Flea and/or animal control; vaccine available for high-risk individuals
Treatment	Streptomycin or ciprofloxacin
Epidemiological Features	United States: endemic in all western and southwestern states; internationally, 95% of human cases occur in Africa, including Madagascar **Category A Bioterrorism Agent**



Tularemia

Tularemia

- Zoonotic disease
- *Francisella tularensis*



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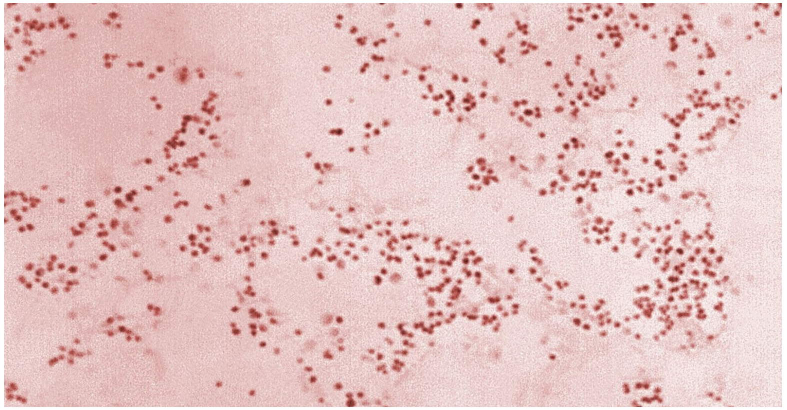
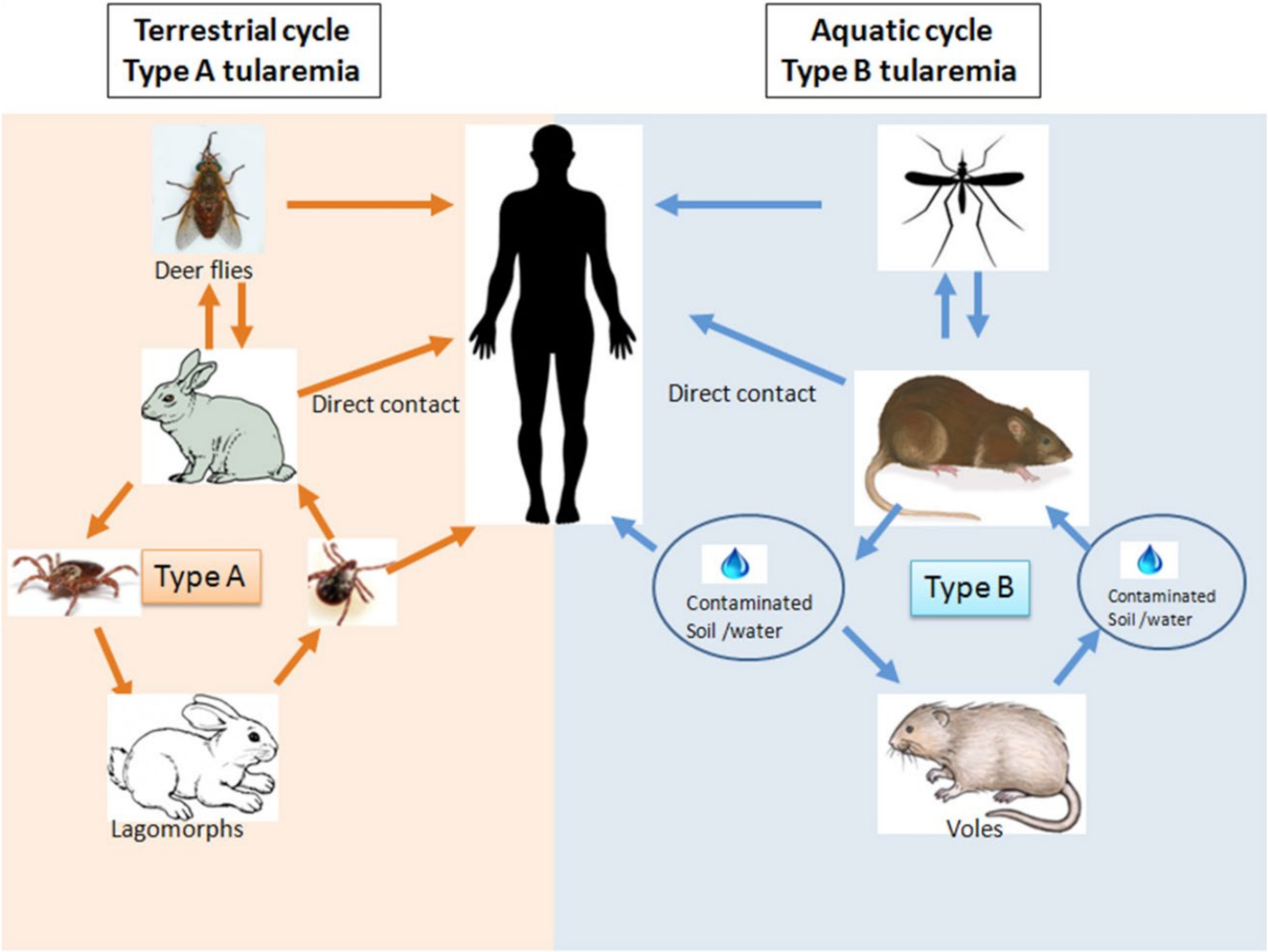
Tularemia

- Zoonotic disease
- *Francisella tularensis*



Gram-negative, facultative intracellular
10-50 required to initiate infection



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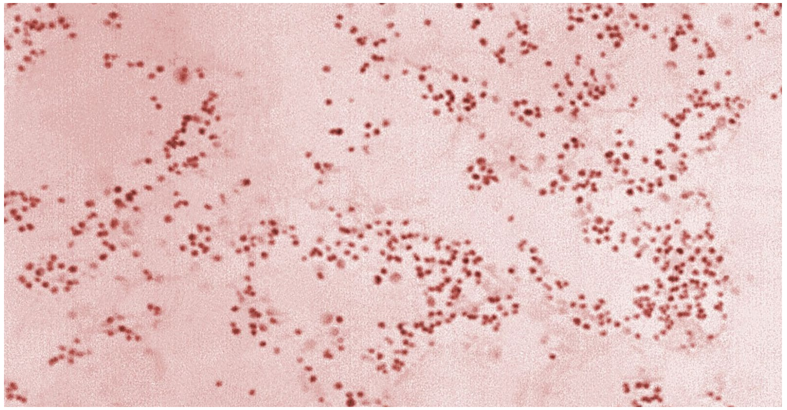
Tularemia

- Zoonotic disease
- *Francisella tularensis*
- Mortality→ No treatment: 30%



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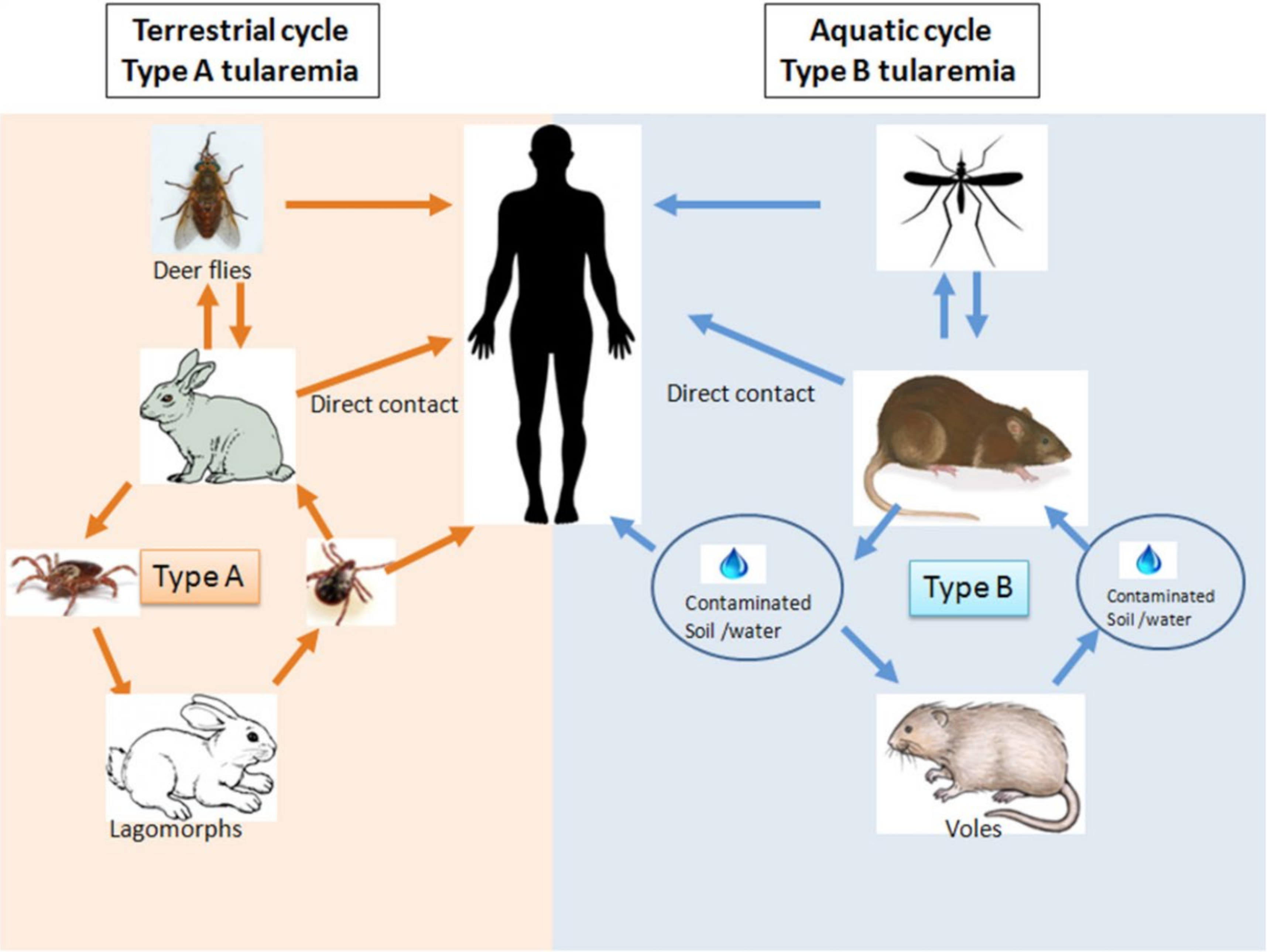


Tularemia

- Zoonotic disease
- *Francisella tularensis*
- Symptoms

•Fever, chills
•Headache
•Malaise, fatigue
•Anorexia
•Myalgia
•Chest discomfort, cough
•Sore throat
•Vomiting, diarrhea
•Abdominal pain

Ulcero (glandular)
•Localized lymphadenopathy
•Cutaneous ulcer at infection site (not always present)

Pneumonic
- Cough (dry or productive)
- Substernal tightness
- Pleuritic chest pain
- Hilar adenopathy, infiltrate, or pleural effusion may be present on chest imaging

Oropharyngeal
- Severe throat pain

Exudative pharyngitis or tonsillitis
- Cervical, preparotid, and/or retropharyngeal lymphadenopathy

Oculoglandular
•Photophobia
•Excessive lacrimation
•Conjunctivitis
•Preauricular, submandibular and cervical lymphadenopathy

Figure 4. A Female Patient With Oculoglandular Tularemia. Patient's picture shows typical appearance of preauricular and submandibular lymphadenomegalia.

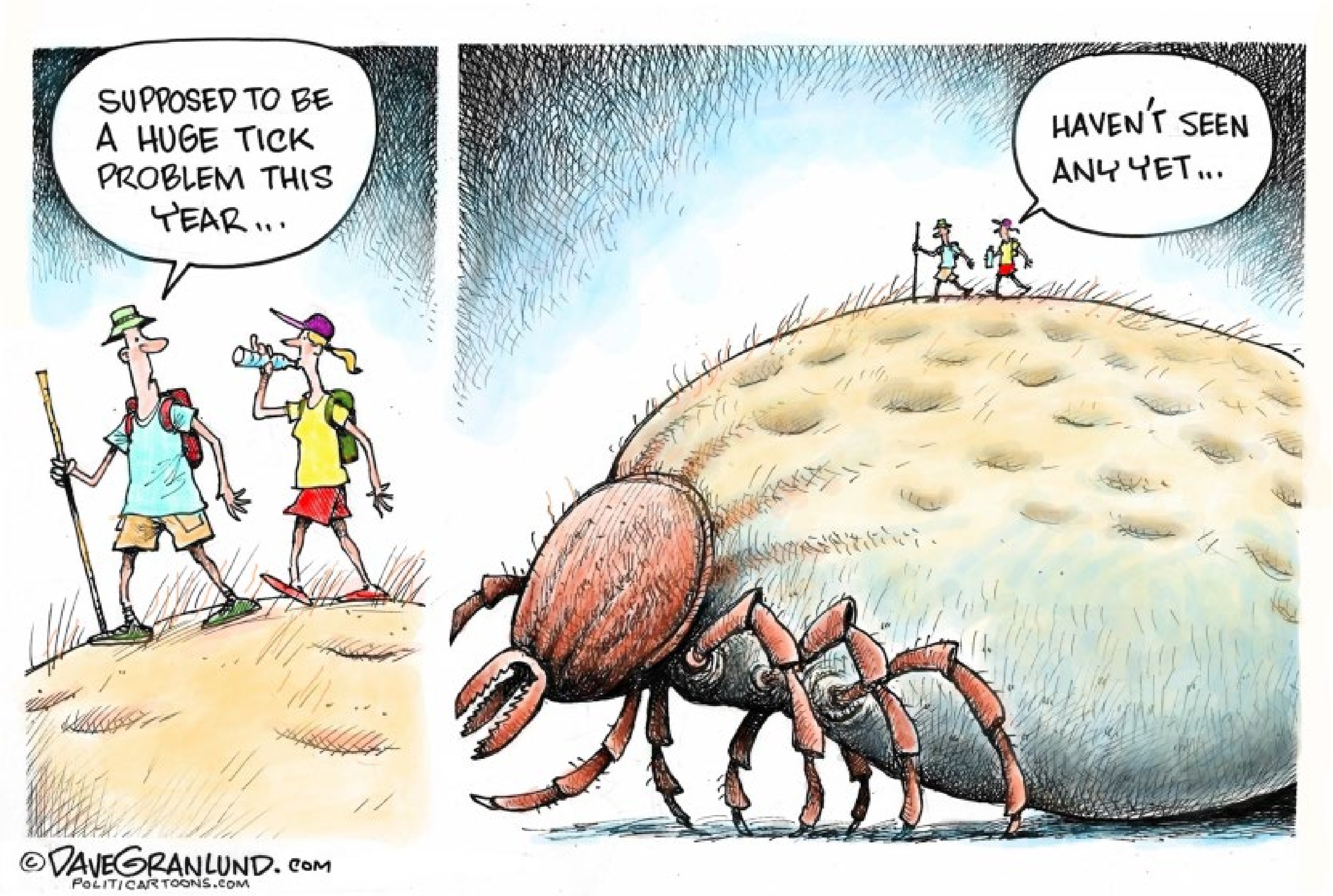
Typhoidal
Characterized by any combination of the general symptoms, without localizing symptoms of other specific presentations

Dinc,G.,Demiraslan,H.,Doganay,M. (2017). 'Unexpected Risks for Campers and Hikers: Tick-Borne Infections', *International Journal of Travel Medicine and Global Health*, 5(1), pp. 5-13. doi: 10.15171/ijtmgh.2017.02

Tularemia Table

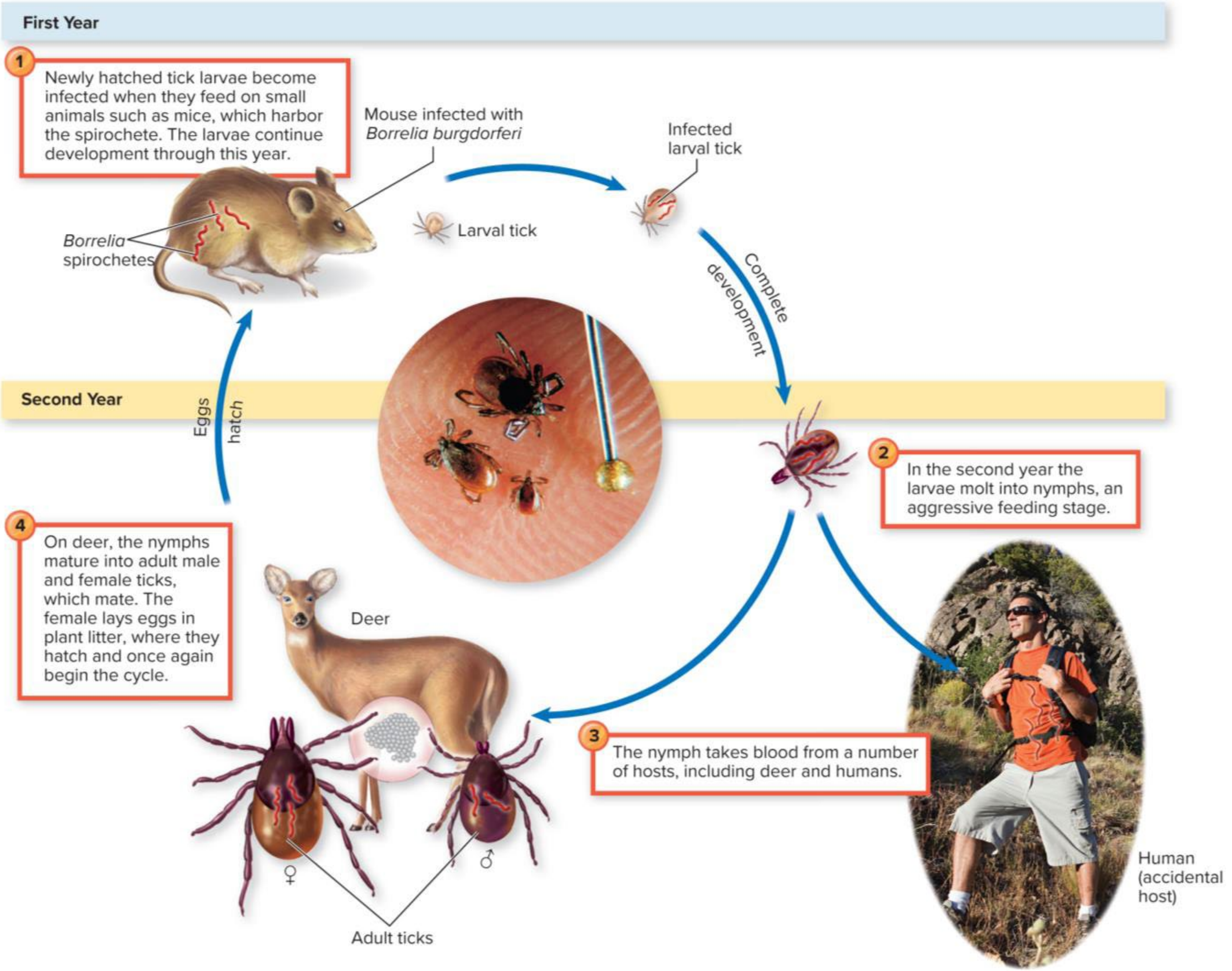
Causative Organism(s)	*Francisella tularensis*
Most Common Modes of Transmission	Biological vector (tick); also direct contact with body fluids from infected animal; airborne
Virulence Factors	Intracellular growth
Culture/Diagnosis	Culture dangerous to lab workers and not reliable; serology most often used; fine needle aspirations of lymph node sometimes used
Prevention	N/A
Treatment	Gentamicin or streptomycin
Epidemiological Features	United States: several hundred cases per year; internationally, 500,000 cases per year **Category A Bioterrorism Agent**

Lyme disease



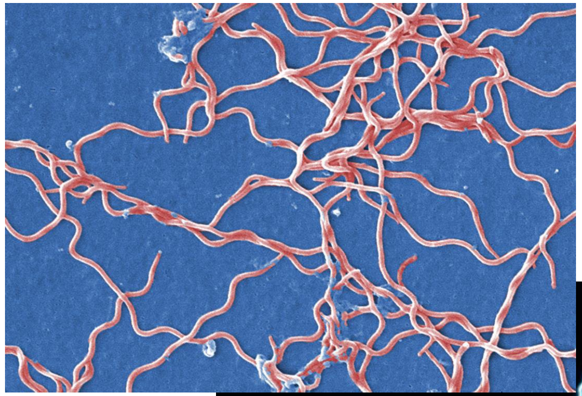
Lyme disease

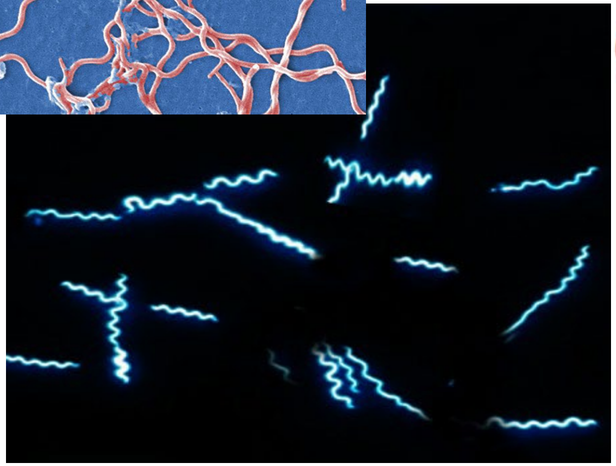
- Zoonotic



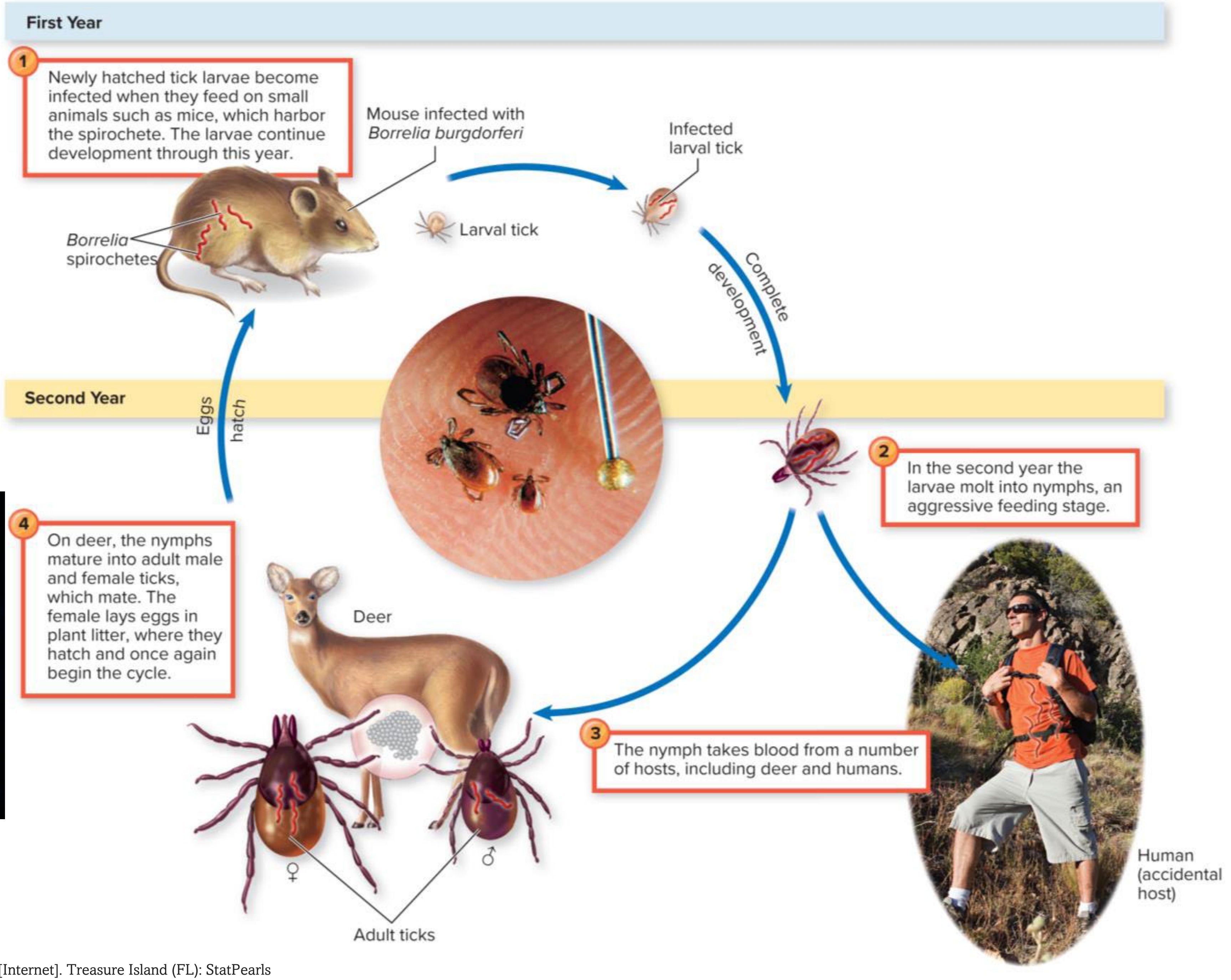
Lyme disease

- Zoonotic





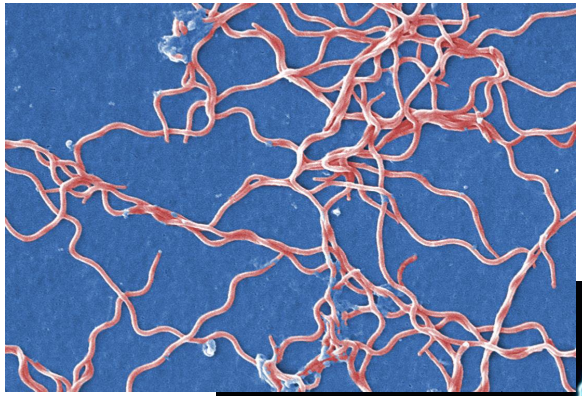
Borrelia burgdorferi
Large spirochete
(with 3 to 10 irregularly spaced coils)

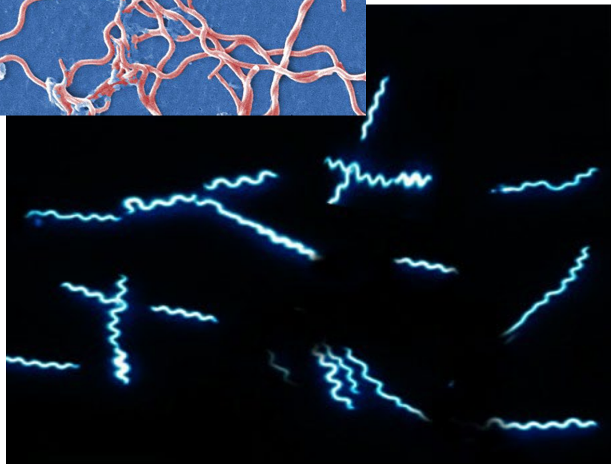


Tatum R, Pearson-Shaver AL. Borrelia Burgdorferi. [Updated 2023 Jul 17]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK532894/

Lyme disease

- Zoonotic





Borrelia burgdorferi
Large spirochete
(with 3 to 10 irregularly spaced coils)

Antigen variation

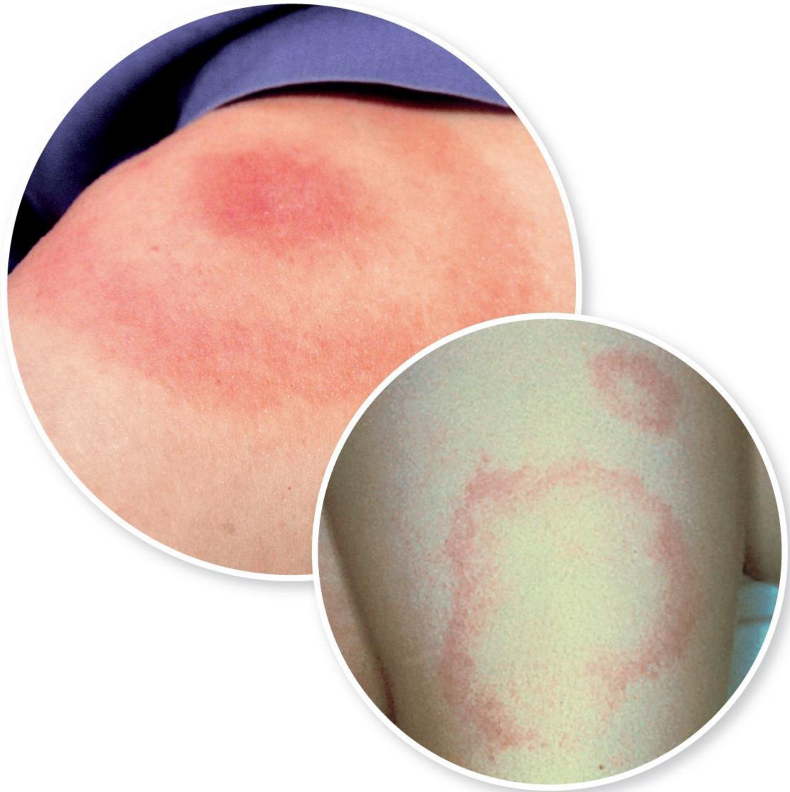
Immune system evasion

Lyme disease pathology

Lyme disease

- Zoonotic
- Symptoms
  - Fever, chills, headache, fatigue, muscle and joint aches, and swollen lymph nodes may occur in the absence of rash
  - Progresses to cardiac and neurological symptoms
  - Develops into crippling polyarthritis after several weeks to months

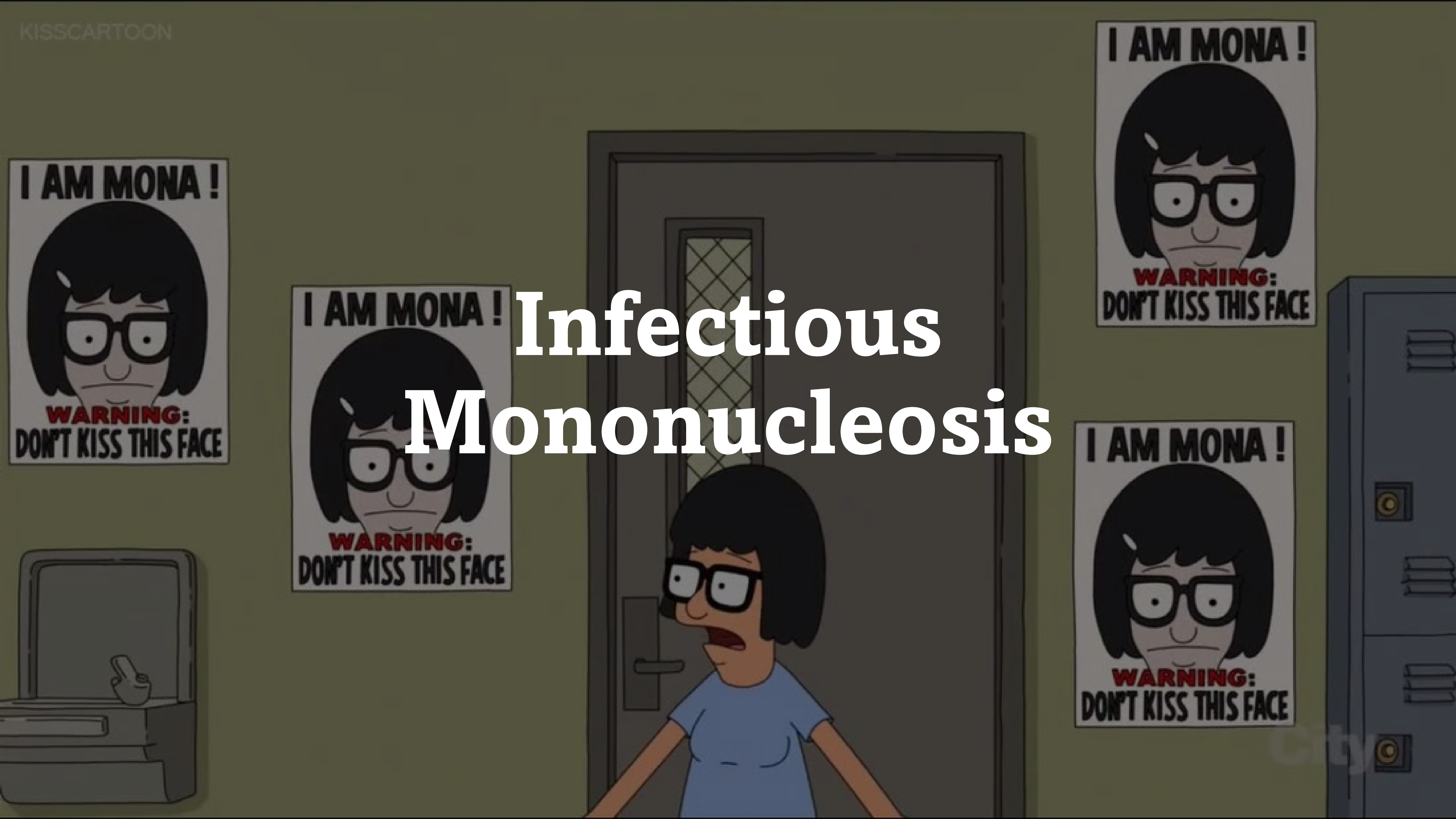
Erythema migrans (EM) rash



Lyme disease Table

Causative Organism(s)	*Borrelia burgdorferi* and closely related species
Most Common Modes of Transmission	Biological vector (tick)
Virulence Factors	Antigenic shifting, adhesins
Culture/Diagnosis	Acute and convelescent sera testing
Prevention	Tick avoidance
Treatment	Doxycycline and/or amoxicillin (3–4 weeks), also cephalosporins and penicillin
Epidemiological Features	In United States, 25,000–30,000 cases per year; endemic in North America, Europe, and Asia

Infectious Mononucleosis

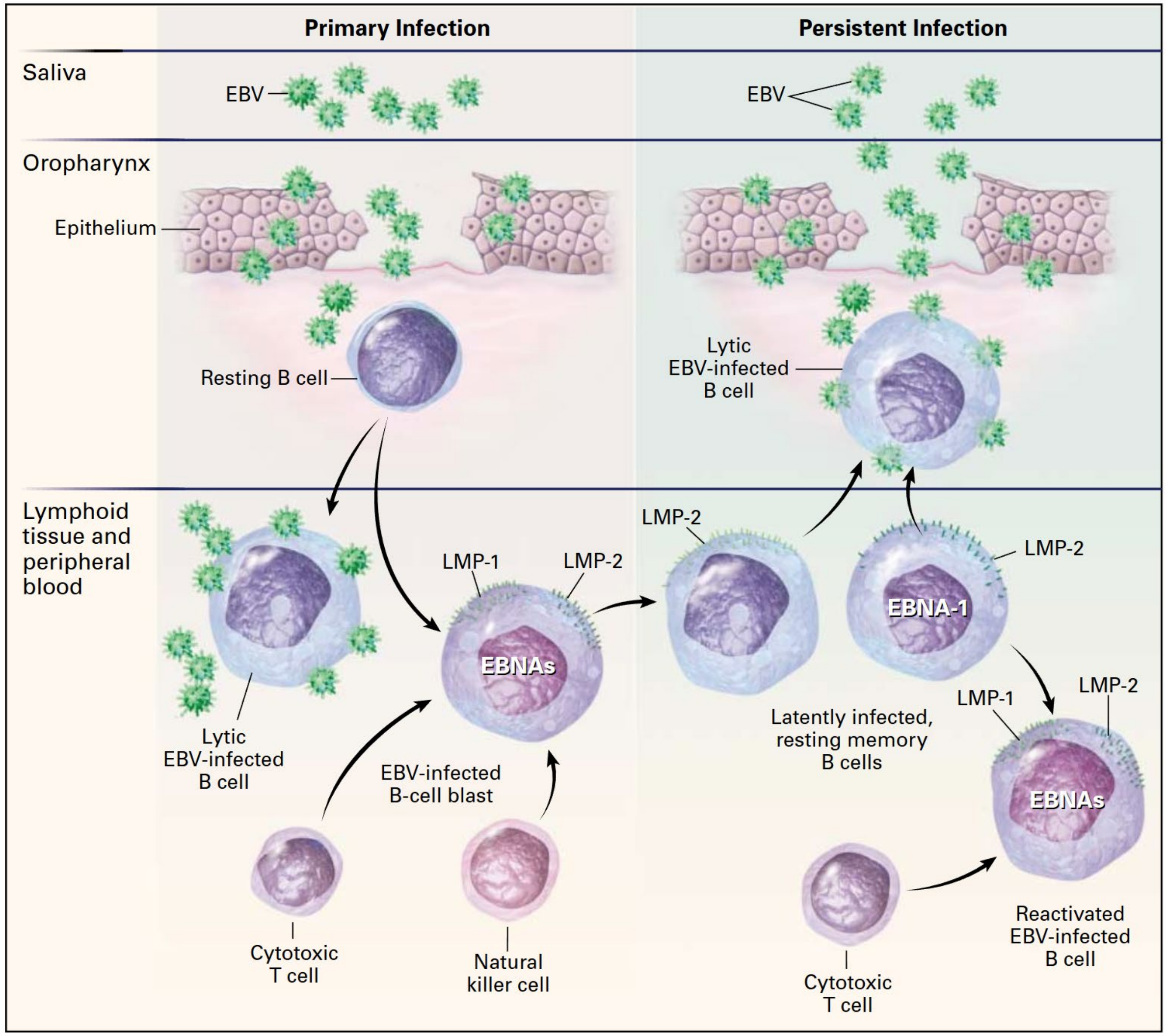


Infections mononucleosis

- Epstein–Barr virus
  - Kissing
  - Sharing drinks and food
  - Sharing drinking cups, eating utensils, or toothbrushes
  - Having contact with toys that children have drooled on
- Symptoms
  - Fatigue
  - Fever
  - Inflamed throat
  - Swollen lymph nodes in the neck
  - Enlarged spleen
  - Swollen liver
  - Rash

90% of the world is infected

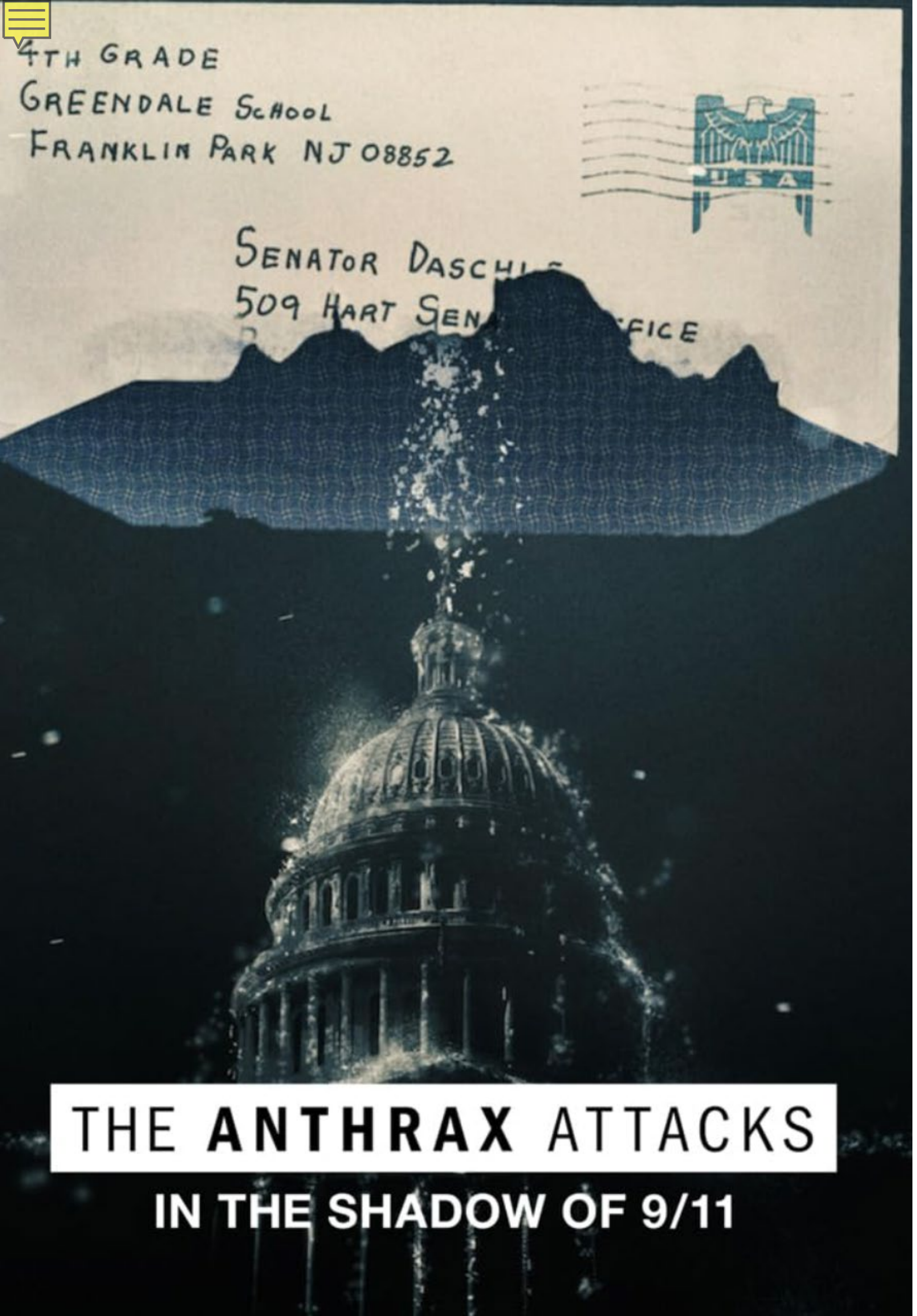
Teens: 30-77% symptomatic



Cohen, J.I., 2000. Epstein–Barr virus infection. *New England journal of medicine*, *343*(7), pp.481-492.

Infectious mononucleosis disease Table

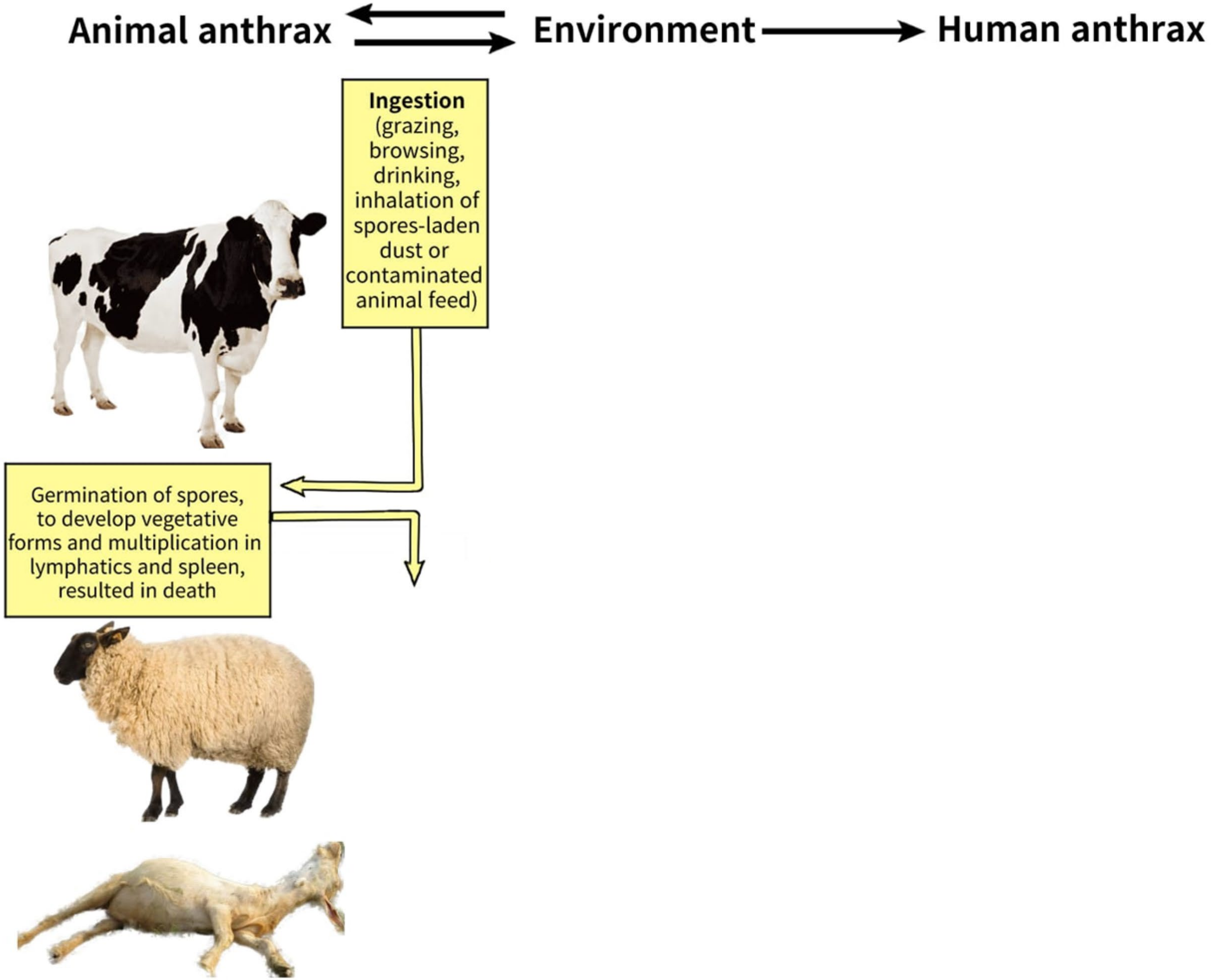
Causative Organism(s)	Epstein–Barr virus (EBV)
Most Common Modes of Transmission	Direct, indirect contact; parenteral
Virulence Factors	Latency, ability to incorporate into host DNA
Culture/Diagnosis	Differential blood count, Monospot test for heterophile antibody, specific ELISA
Prevention	N/A
Treatment	Supportive
Distinctive Features	Lifelong persistence
Epidemiological Features	United States: 500 cases per 100,000 per year


4TH GRADE
GREENDALE SCHOOL
FRANKLIN PARK NJ 08852
USA
THE ANTHRAX ATTACKS
IN THE SHADOW OF 9/11

Anthrax

Anthrax

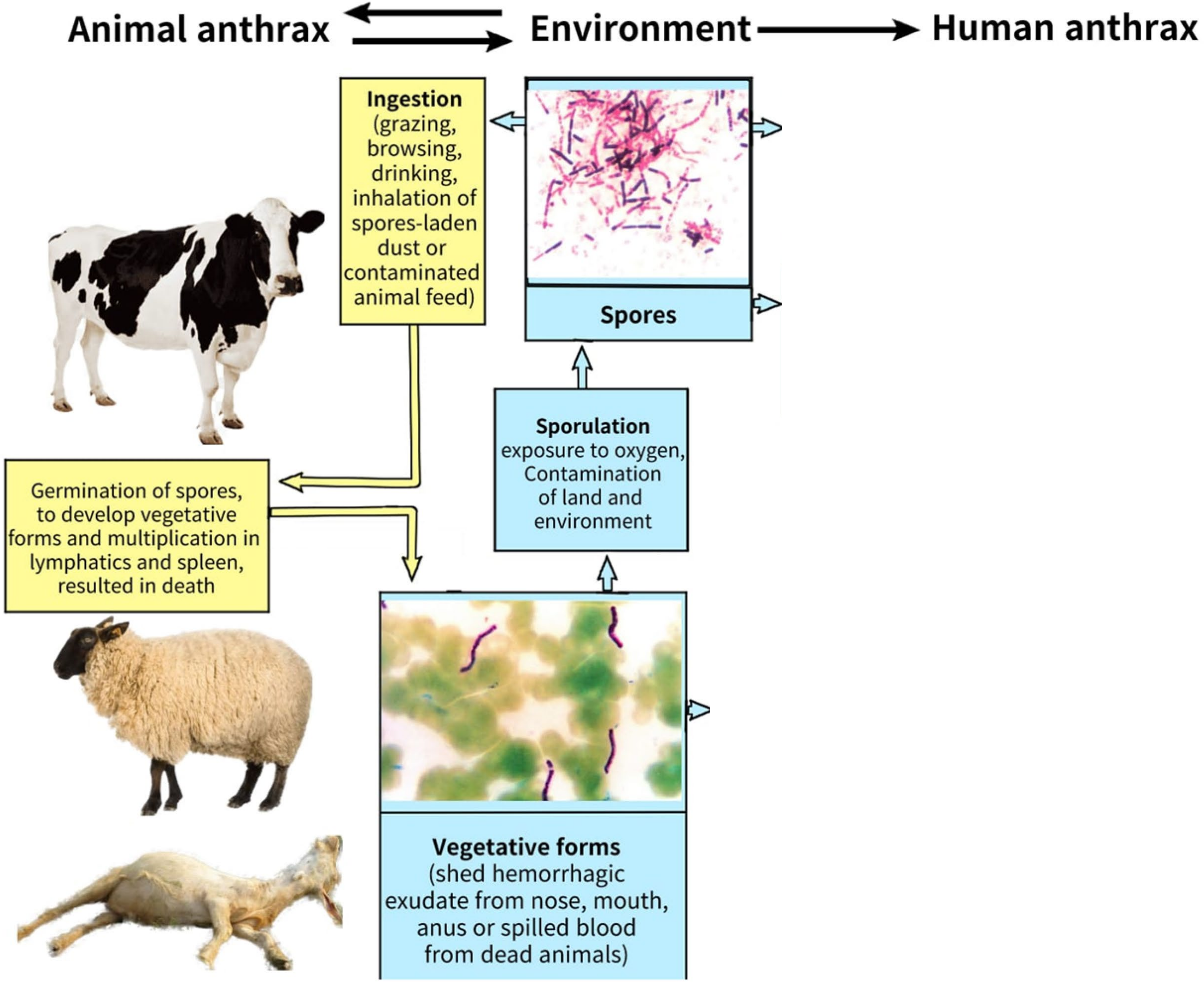
- Zoonotic



Doganay, M., Dinc, G., Kutmanova, A., & Baillie, L. (2023). Human Anthrax: Update of the Diagnosis and Treatment. *Diagnostics*, *13*(6), 1056. https://doi.org/10.3390/diagnostics13061056

Anthrax

- Zoonotic



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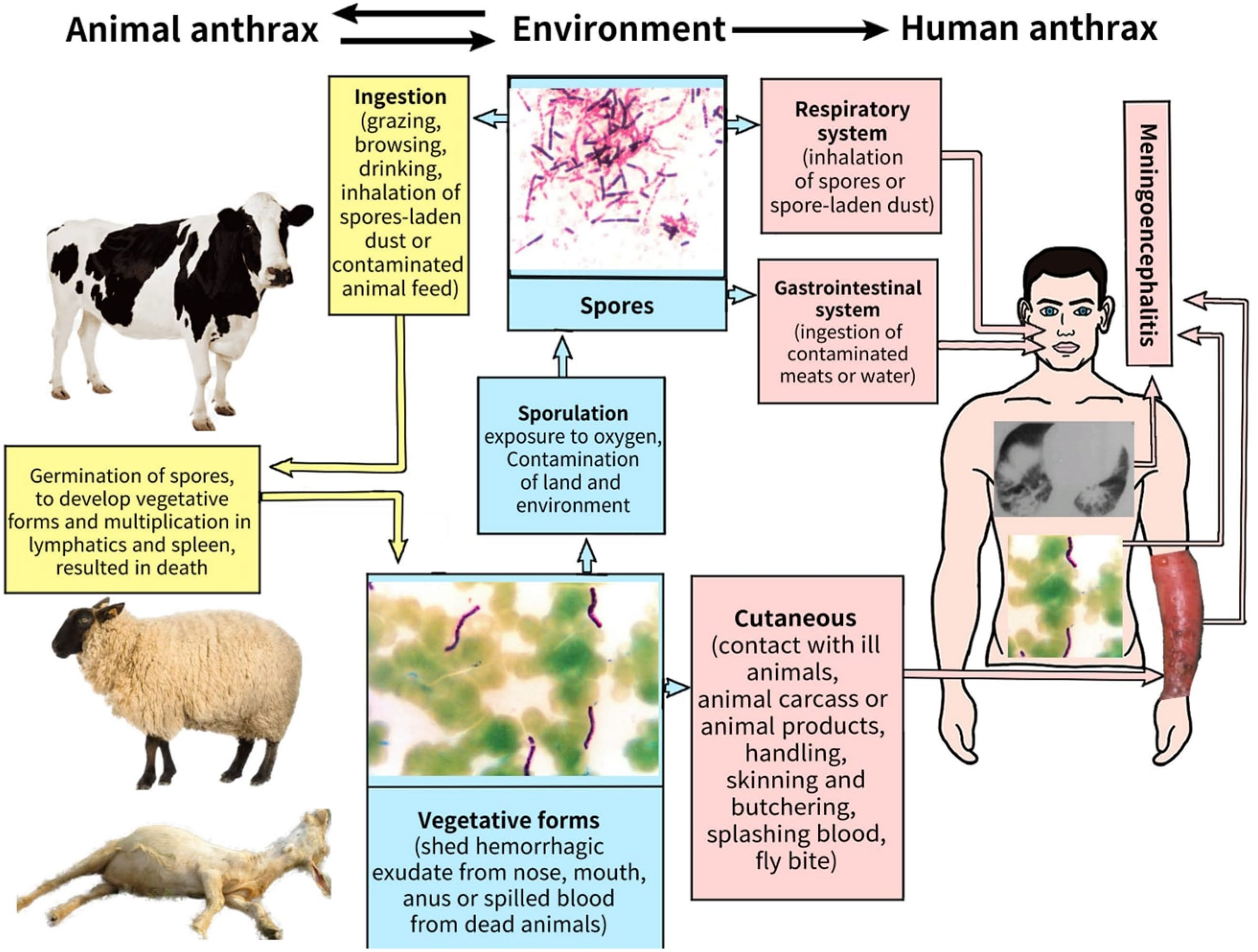
Anthrax

- Zoonotic



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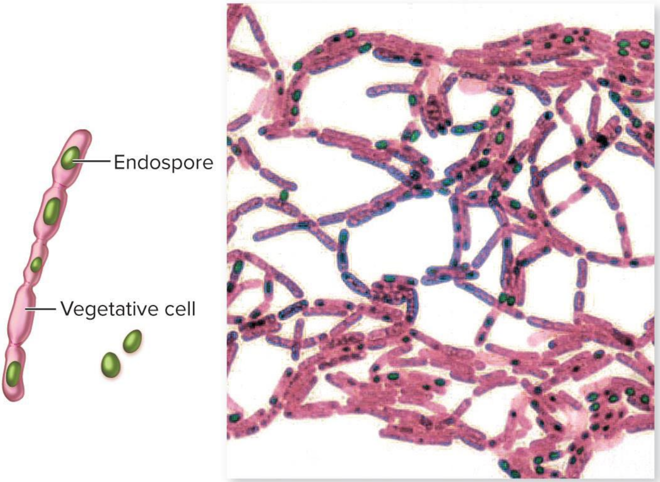
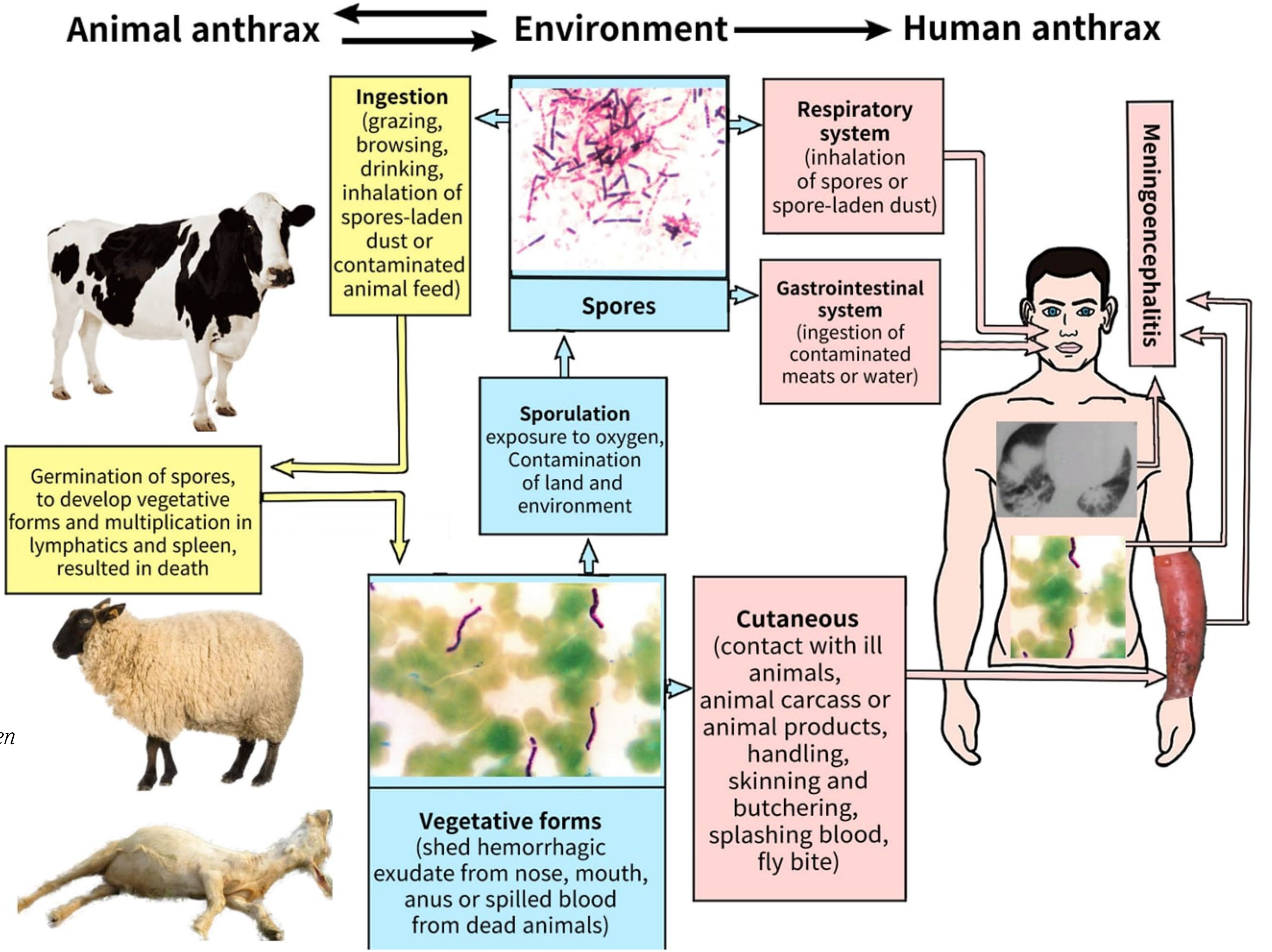
Anthrax

- Zoonotic



Bacillus anthracis
Gram-positive rod
Aerobic
Catalase positive

Tripartite toxin
Edema factor
Protective antigen
Lethal factor



Doganay, M., Dinc, G., Kutmanova, A., & Baillie, L. (2023). Human Anthrax: Update of the Diagnosis and Treatment. *Diagnostics*, *13*(6), 1056. https://doi.org/10.3390/diagnostics13061056

Anthrax

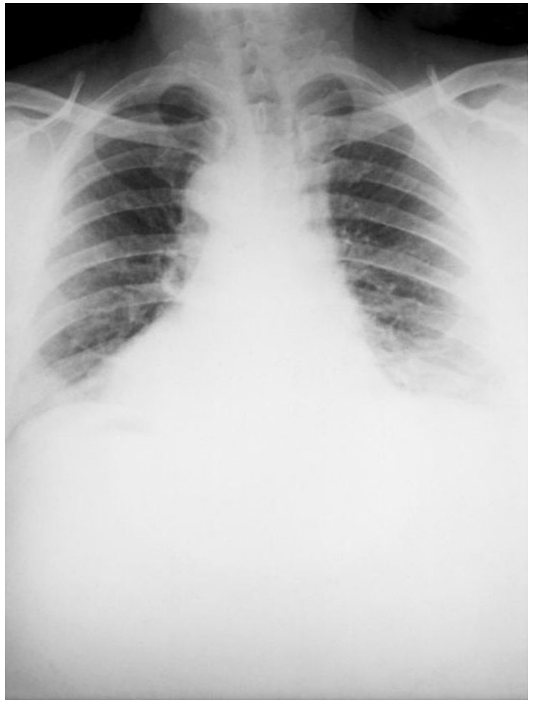
- Symptoms

Cutaneous anthrax
•Group of itchy, small blisters or bumps
•Large amount of swelling around the sore
•Painless sore commonly on the face, neck, arms, or hand that has a black center
  - Sore appears after the bumps have gone down

•For injection anthrax, infected sores (abscesses) deep under the skin or muscle at the injection site

Gastrointestinal anthrax
•Fever and chills
•Swelling of neck or neck glands
•Sore throat, hoarseness, and pain when swallowing
•Nausea and vomiting, especially bloody vomiting
•Diarrhea or bloody diarrhea
•Headache
•Red face and red eyes
•Stomach pain and swelling
•Fainting

Inhalation anthrax
•Fever and chills
•Heavy sweats
•Chest pain, cough, or shortness of breath
•Confusion or dizziness
•Nausea, vomiting, or stomach pains
•Headache or body aches
•Extreme tiredness



Anthrax disease Table

Causative Organism(s)	*Bacillus anthracis*
Common Modes of Transmission	Vehicle (air, soil), indirect contact (animal hides), vehicle (food)
Virulence Factors	Triple exotoxin
Culture/Diagnosis	Culture, direct fluorescent antibody tests
Prevention	Vaccine for high-risk population; used in conjunction with antibiotics post-exposure
Treatment	In consultation with the CDC
Epidemiological Features	Internationally, 2,000–20,000 cases annually, most cutaneous **Category A Bioterrorism Agent**

Hemorrhagic Fevers

- Agents that infect the blood and lymphatics
- Extreme fevers accompanied by internal hemorrhaging
- RNA enveloped viruses
- Aedes genus of mosquito
- Prevalence fluctuates due to global warming patterns

Nonhemorrhagic Fevers

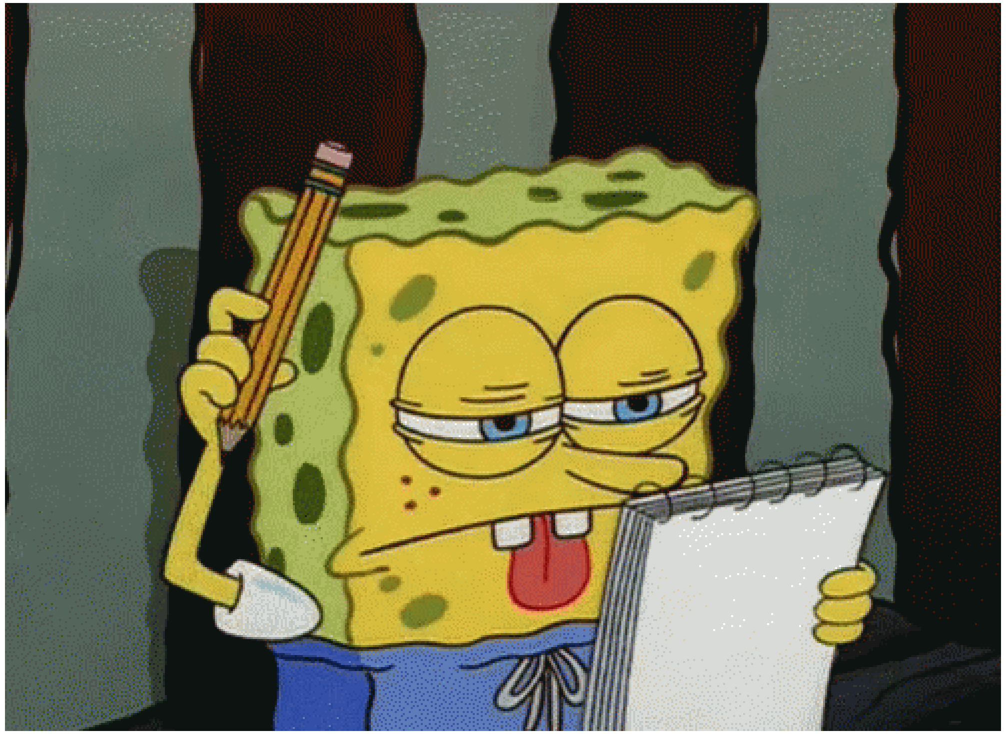
- High fever but without capillary fragility leading to hemorrhagic symptoms
- Mostly caused by multiple bacterial species and one protozoan (Babesia) species

Group activity

- Each group will be responsible for one disease and must answer:
    1. Is it a hemorrhagic or nonhemorrhagic fever disease?
    2. What is the causative agent?
    3. What is the mode of transmission?
    4. What are the main symptoms?
    5. In what regions or countries is this disease considered endemic?

Group activity

- Swap your notes with another team and give each other feedback
  - Did they answer all the questions correctly?
  - Were the answers specific or vague?
  - What could be improved?

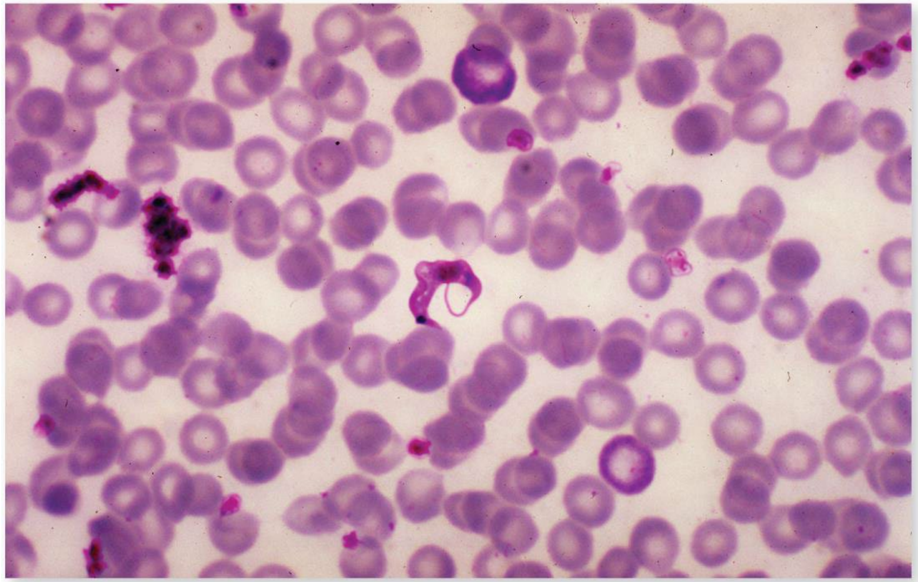


Chagas disease

Chagas disease

- Vector:

Triatomine Bugs



Trypanosoma cruzi
Flagellated protozoan, similar to the agent of African sleeping sickness

Chagas disease

- Symptoms

Romaña's sign

Acute phase
•Fever
•Feeling tired
•Body aches
•Headache
•Rash
•Loss of appetite
•Diarrhea
•Vomiting
•Eyelid swelling (Romaña's sign)

Chronic phase
•Heart issues, such as an enlarged heart, heart failure, altered heart rate or rhythm, or sudden death.
•Digestive problems, such as an enlarged esophagus or colon, leading to trouble eating or going to the bathroom.

Chagas disease Table

Causative Organism(s)	*Trypanosoma cruzi*
Most Common Modes of Transmission	Biological vector (triatomine bug), vertical
Virulence Factors	Antioxidant enzymes, co-opting host antigens; induces autoimmunity
Culture/Diagnosis	Blood smear in acute phase; serological methods in later stages
Prevention	Insect control
Treatment	Consult CDC
Epidemiological Features	Endemic in Central and South America; 230,000 cases present in the United States; considered a **neglected parasitic infection (NPI)**

Malaria

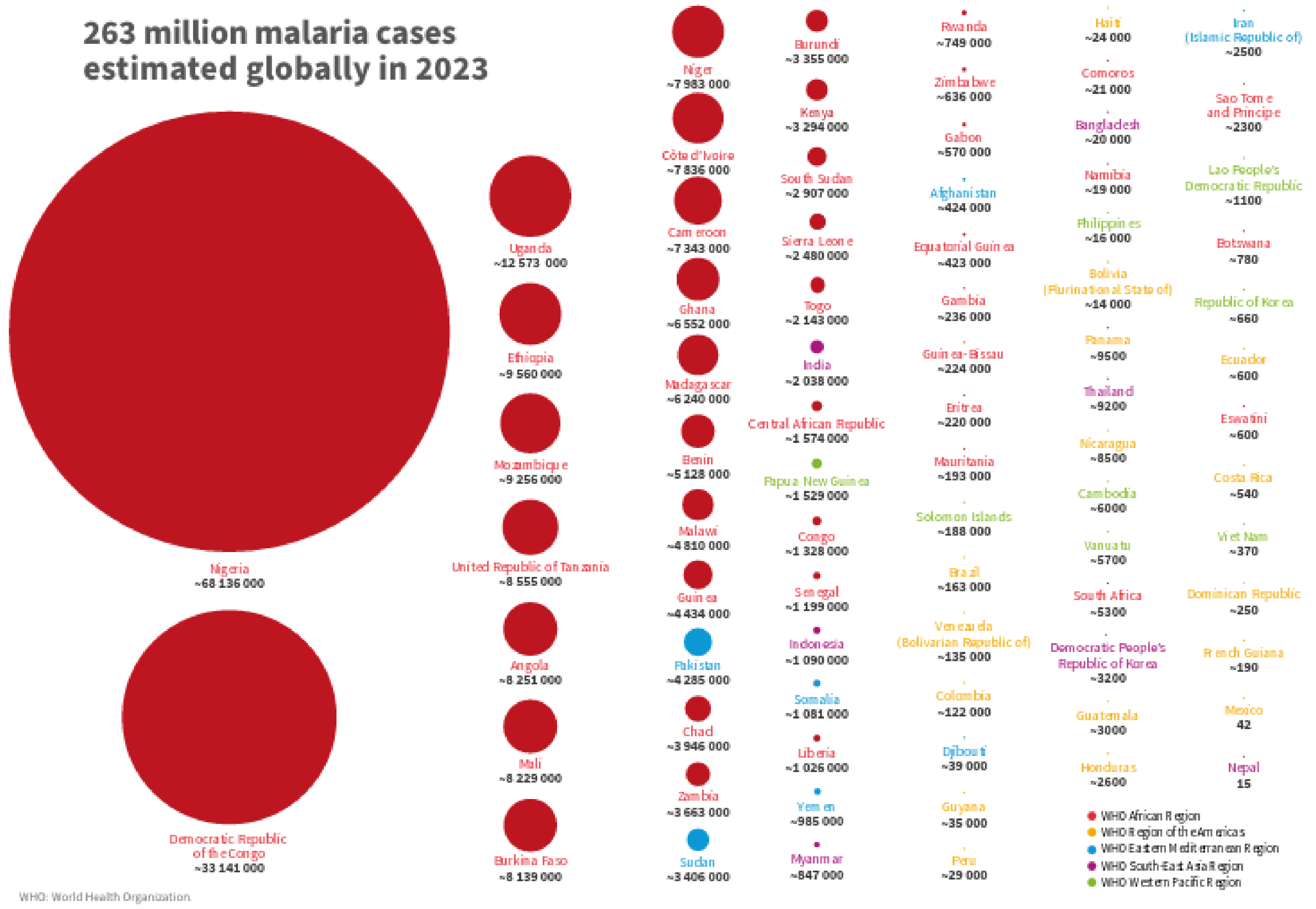
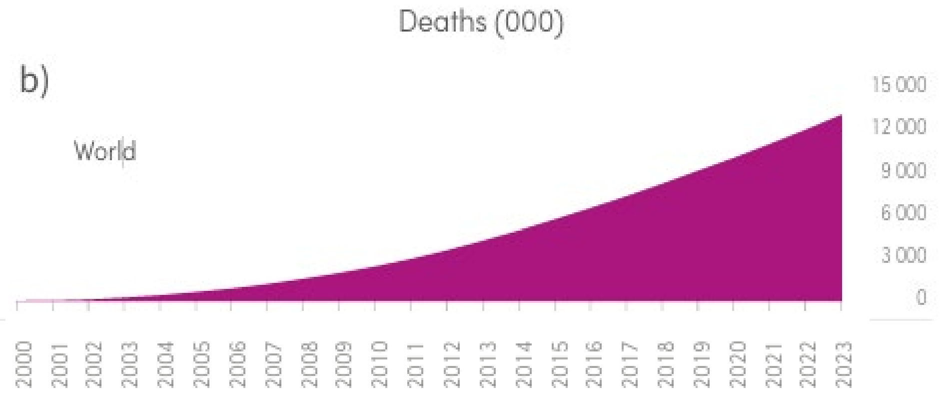
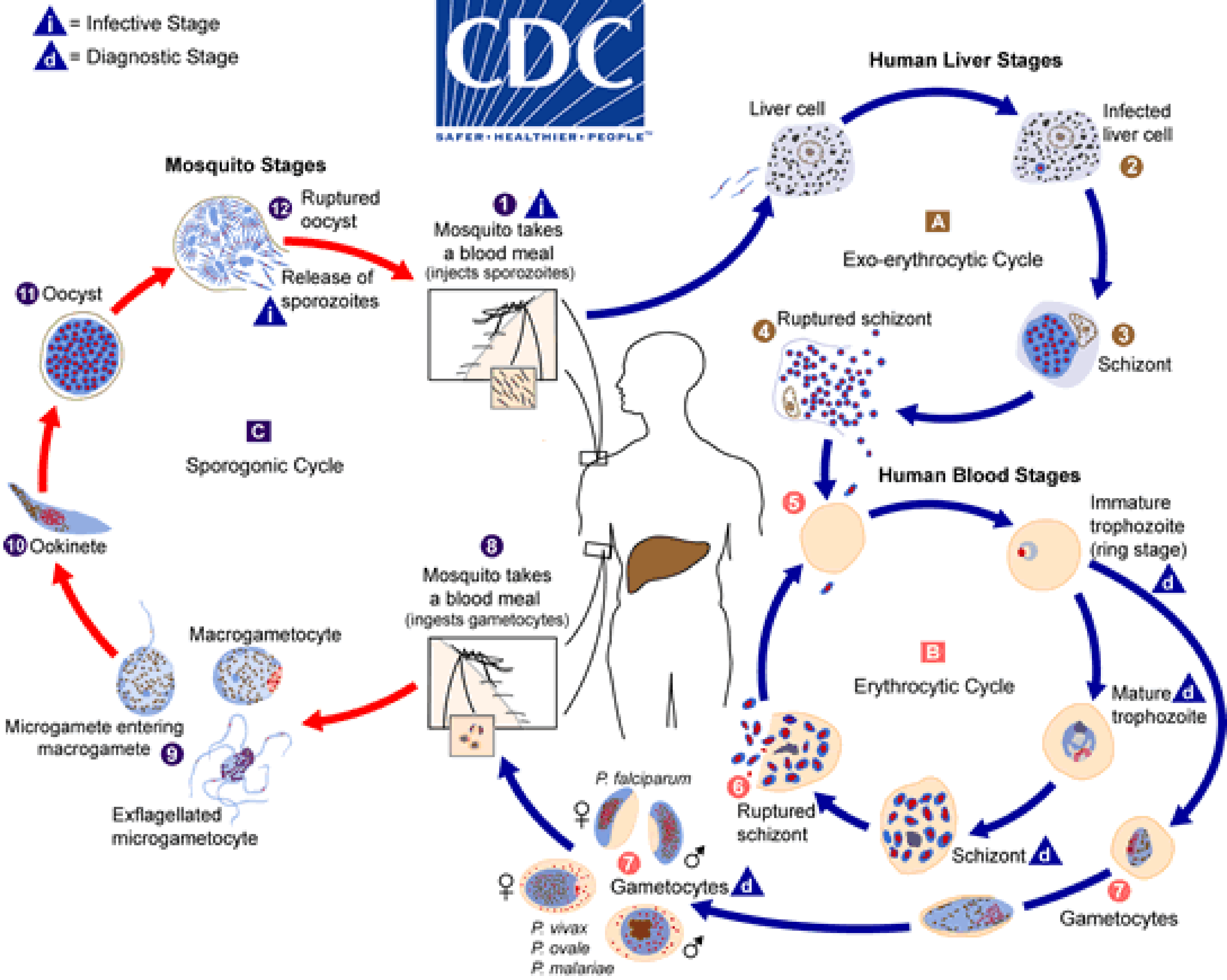


Fig. 2.1. Estimated number of malaria cases per country and area in 2023 *Source: WHO database.*



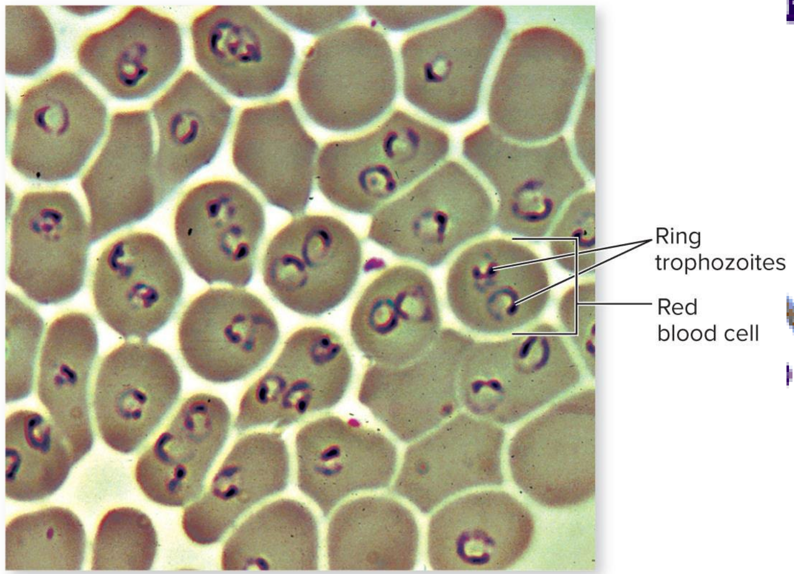
Malaria

- Vector: Anopheles



Malaria

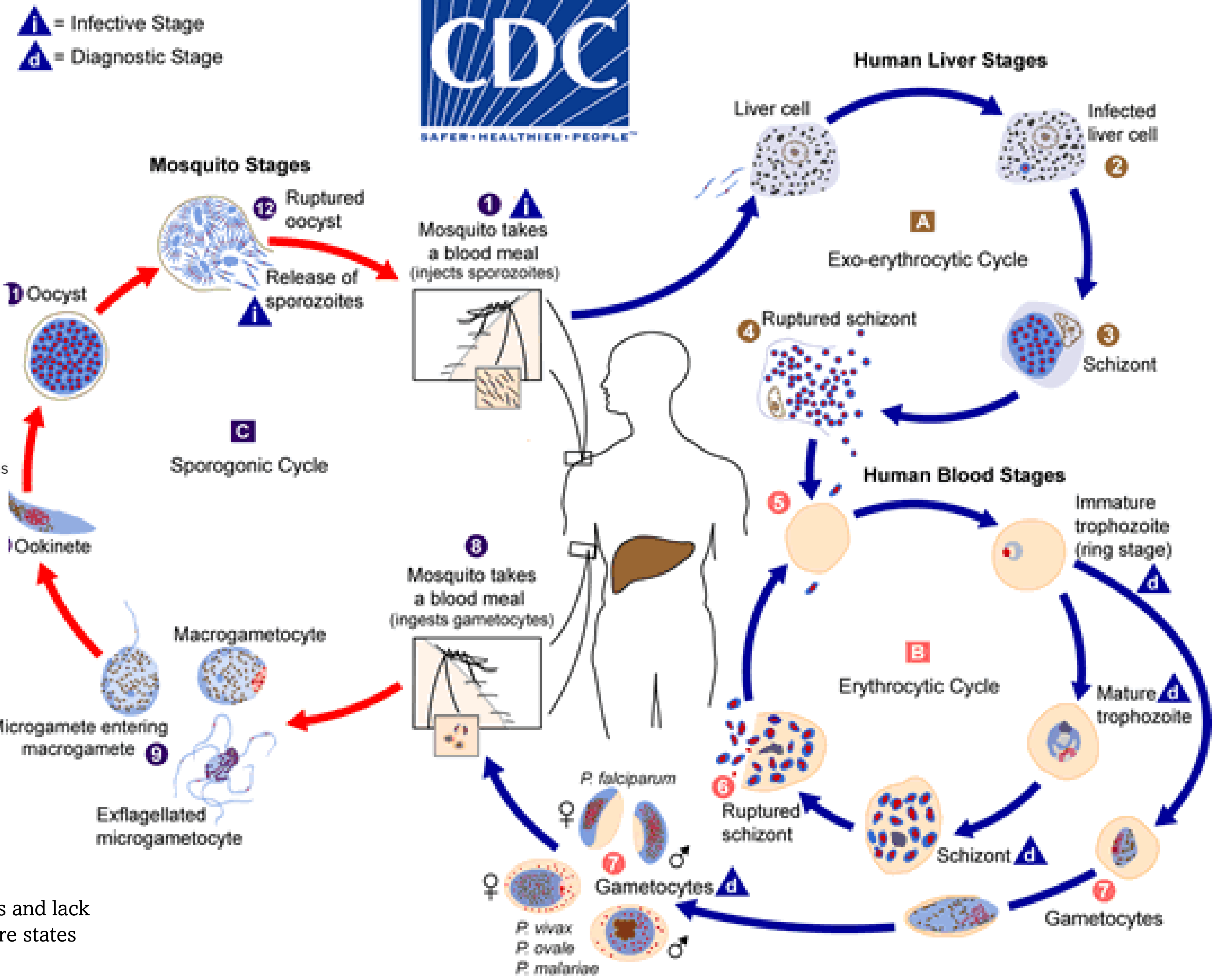
- Vector: Anopheles



Plasmodium

P. malariae
P. vivax
P. falciparum
P. ovale
P. knowlesi

Apicomplexans: live in animal hosts and lack locomotor appendages in the mature states



Malaria

- Symptoms

•Fever and flu-like illness

•Chills

•Headache, muscle aches, and tiredness

•Nausea, vomiting, and diarrhea may also occur

•Kidney failure

•Seizures

•Mental confusion

•Coma

- Prevention

Long-term mosquito abatement

- Elimination of standing water
- Spraying of insecticides
- Introducing sterile male mosquitoes into populations
- Mosquito nets, screens, repellants

Human chemoprophylaxis

- Prophylactic treatment for travelers
- A vaccine shows promise and is recommended for children in endemic areas

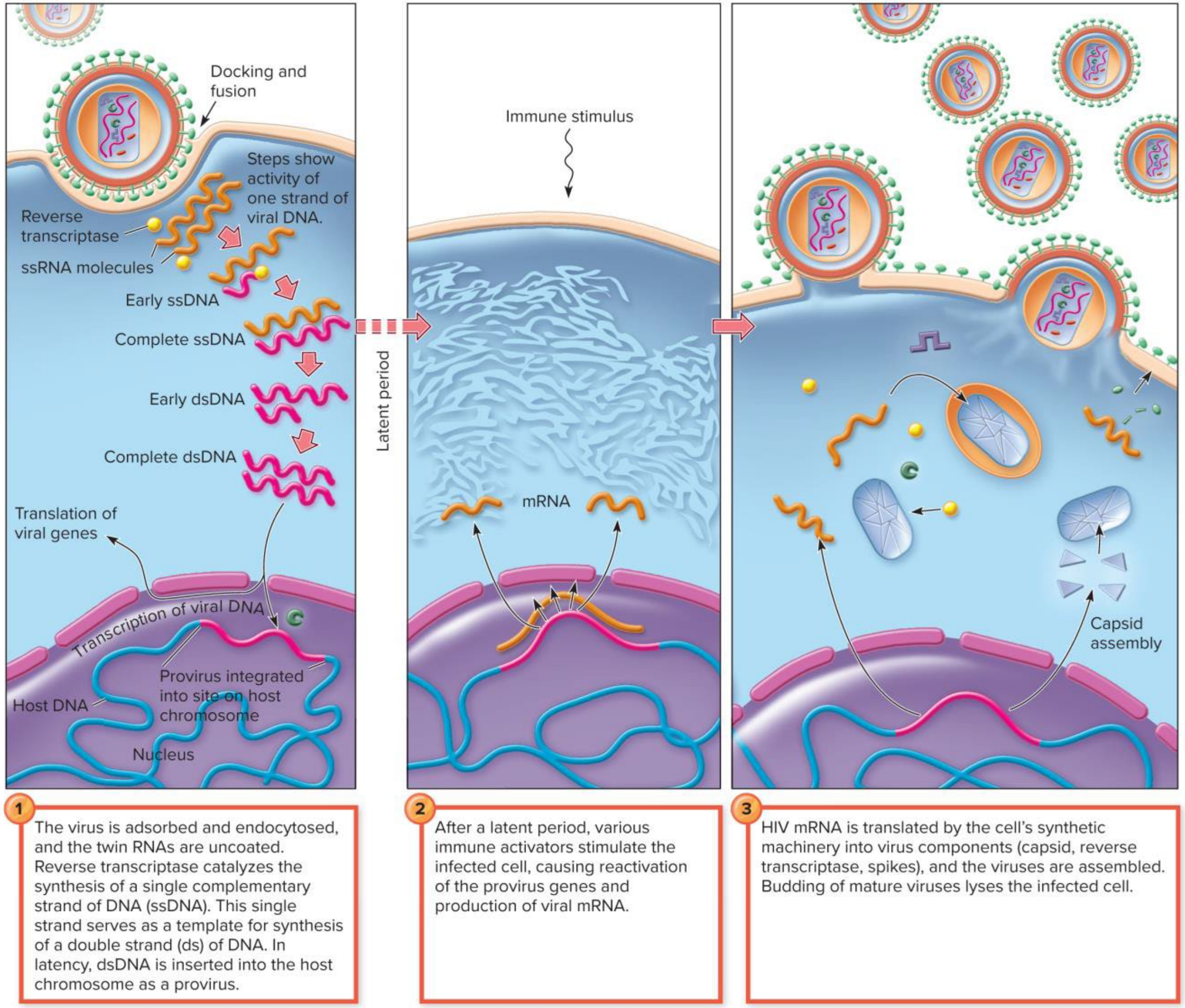
Malaria disease Table

Causative Organism(s)	*Plasmodium falciparum, P. vivax, P. ovale, P. malariae, P. knowlesi*
Most Common Modes of Transmission	Biological vector (mosquito), vertical
Virulence Factors	Multiple life stages; multiple antigenic types, ability to scavenge glucose, GPI toxin, cytoadherence
Culture/Diagnosis	Blood smear; serological methods
Prevention	Mosquito control; use of bed nets; for children in endemic areas now beginning use of RTS, S vaccine; prophylactic antiprotozoal agents
Treatment	Artemisinin, combination therapy; consult WHO
Epidemiological Features	United States: cases are generally in travelers or immigrants; internationally, 300 million cases in “malaria belt”; half million deaths per year; more deadly in children

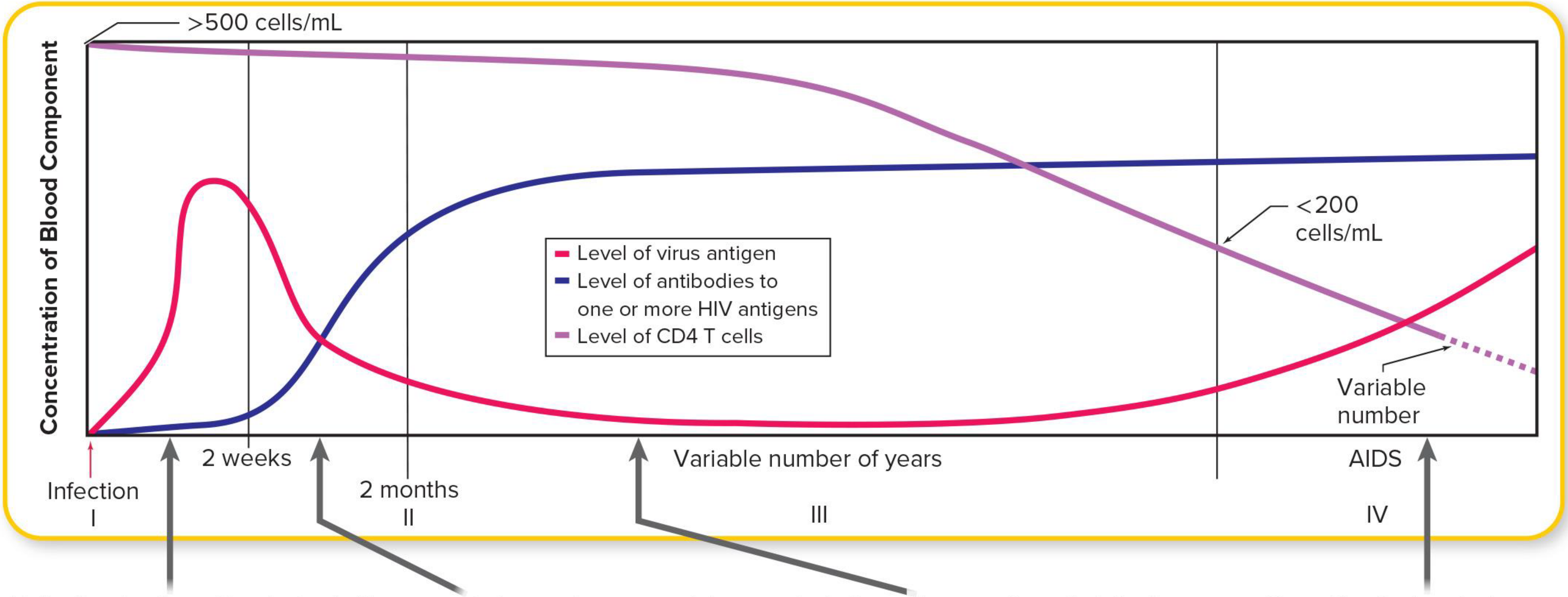
Human Immunodeficiency Virus

Human Immunodeficiency Virus (HIV)

- HIV-1: most dominant form in the world (Most related to simian immunodeficiency viruses in chimpanzees)
- HIV-2
- Infects T-cells (also macrophages)



Human Immunodeficiency Virus (HIV)



Initial infection is often attended by vague, mononucleosis-like symptoms that soon disappear. This phase corresponds to the initial high levels of virus (the green line above). Antibodies are not yet abundant.

In the second phase, virus numbers in blood drop dramatically and antibody begins to appear. CD4 T cells begin to decrease in number.

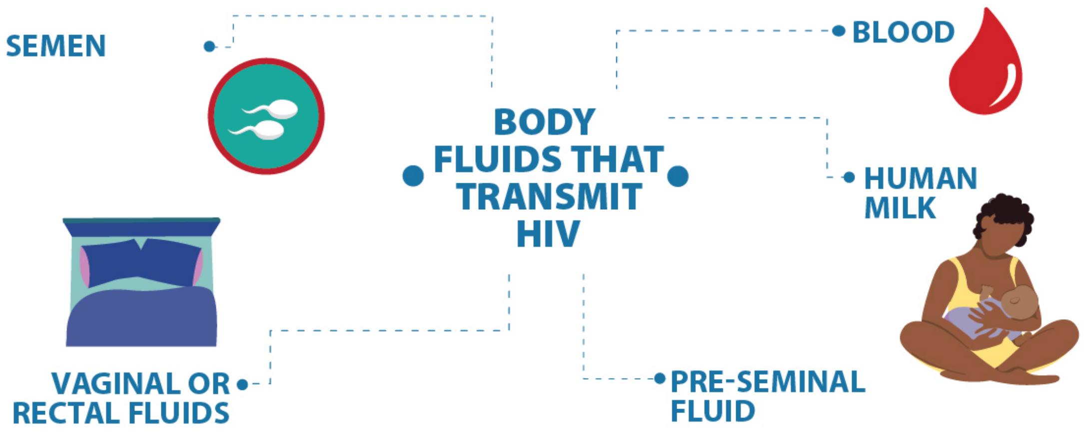
A long period of mostly asymptomatic infection ensues. During this time, which can last from 2 to 15 years, swollen lymph nodes may be the prominent symptom. During the mid- to late-asymptomatic period, the number of T cells in the blood is steadily decreasing. Once the T-cell level reaches a (low) threshold, the symptoms of AIDS ensue.

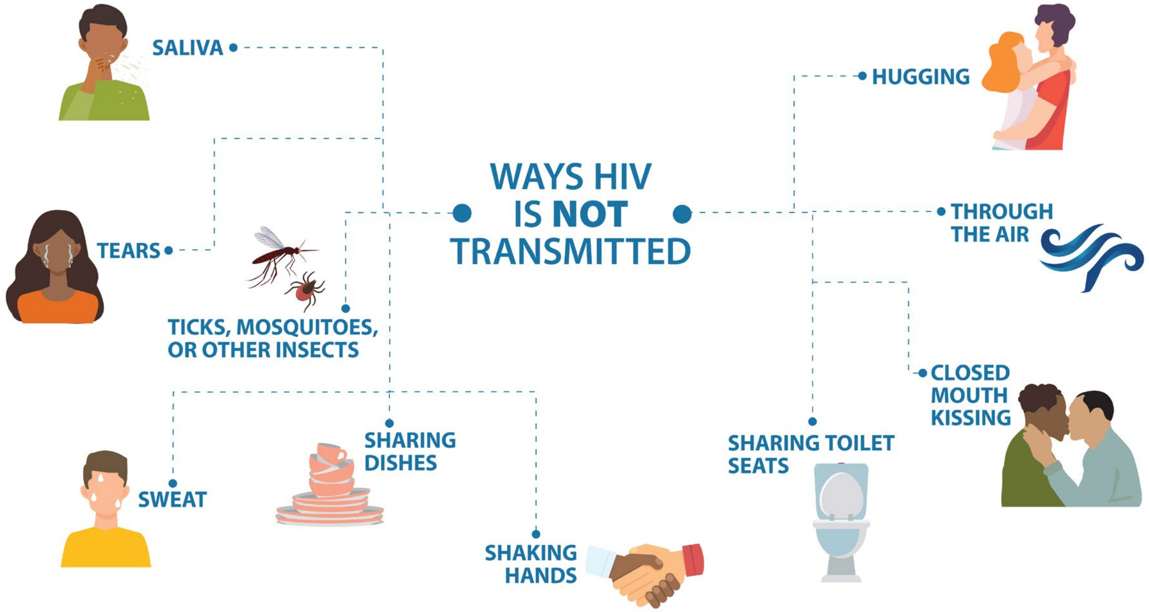
Once T cells drop below 200 cells/mL, AIDS results. Note that even though antibody levels remain high, virus levels in the blood begin to rise.

Human Immunodeficiency Virus (HIV)

Acquired Immune Deficiency Syndrome (AIDS)

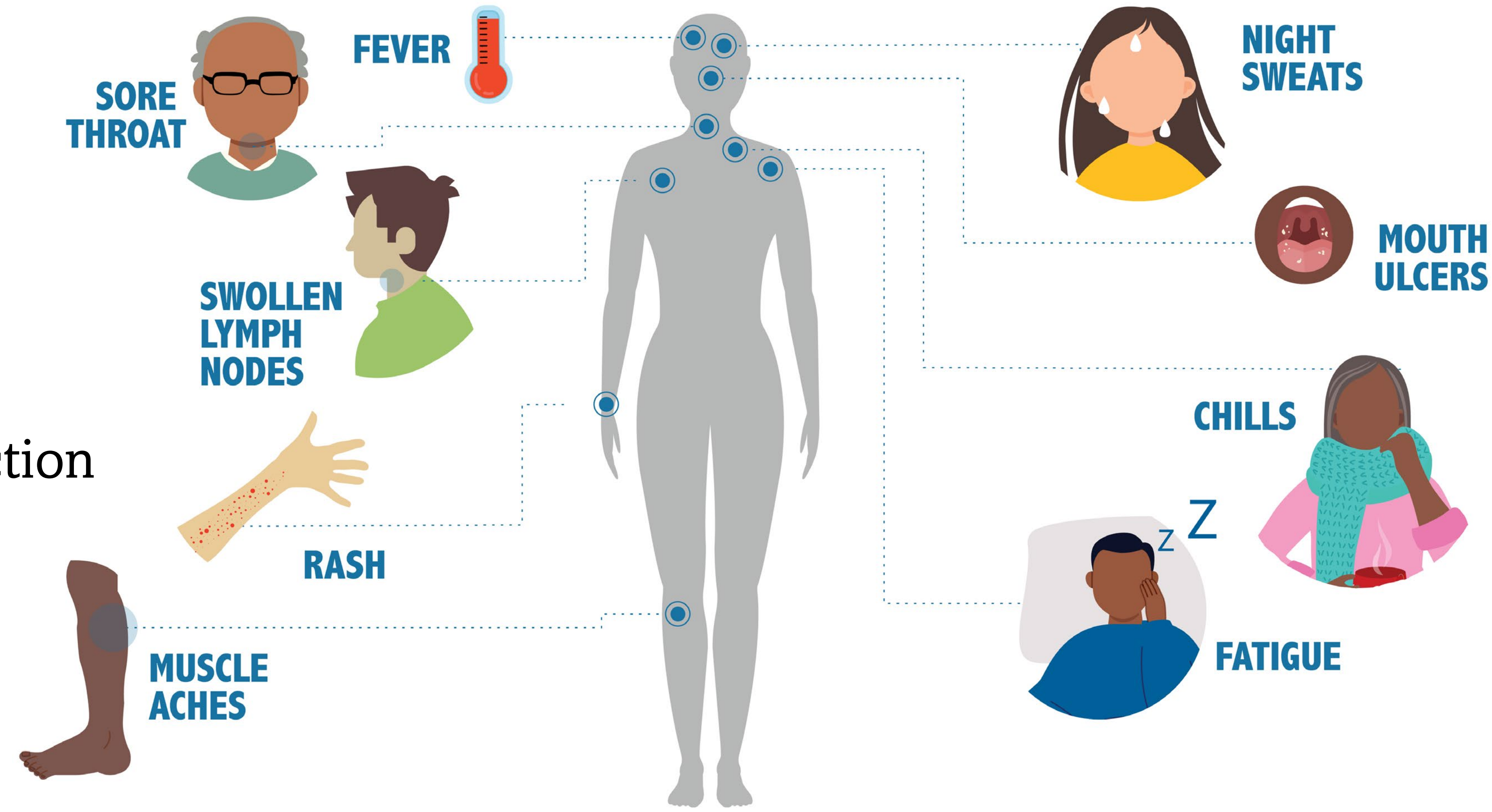
Only **certain body fluids** from a person who has HIV **can transmit HIV**.





Human Immunodeficiency Virus (HIV)

- Symptoms
- Pneumonia caused by *Pneumocystitis jiroveci*
- Kaposi's sarcoma
- Sudden weight loss
- Swollen lymph nodes
- General loss of immune function



Human Immunodeficiency Virus (HIV)

- Epidemiology
- 630 000 people died of HIV-related illnesses worldwide in 2024

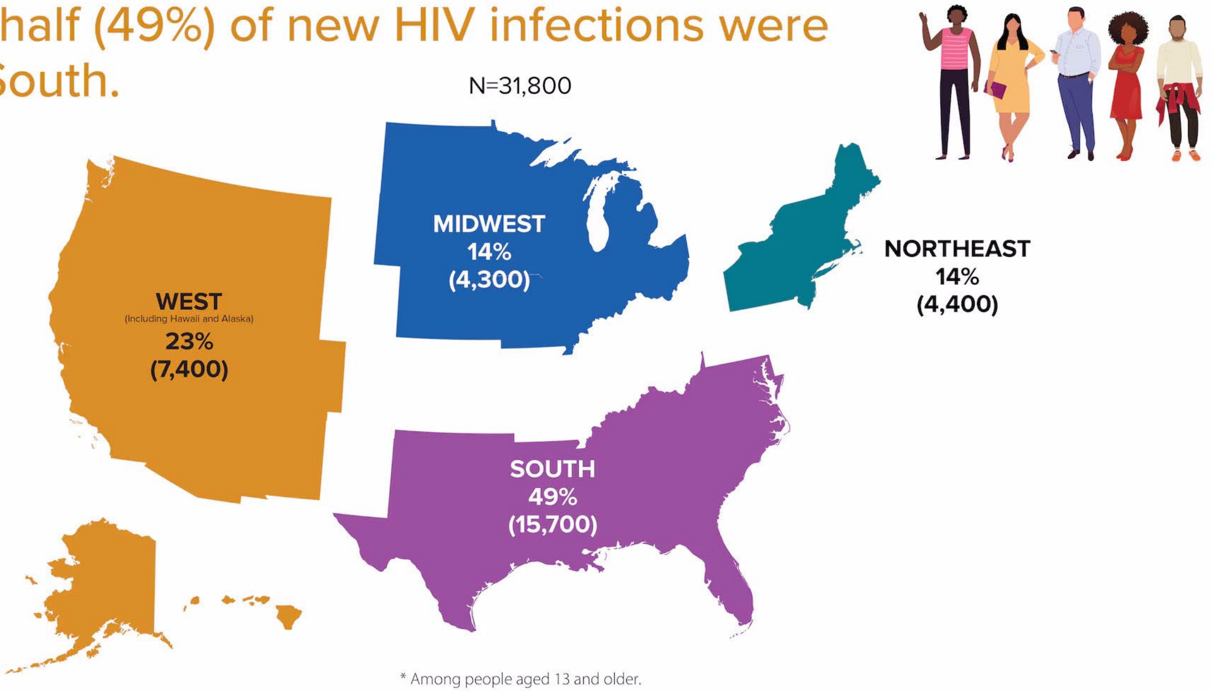
Summary of the global HIV epidemic, 2024

	People living with HIV	People acquiring HIV	People dying from HIV-related causes
Total	**40.8 million** [37.0–45.6 million]	**1.3 million** [1.0–1.7 million]	**630 000** [490 000–820 000]
Adults (15+ years)	**39.4 million** [35.7–44.0 million]	**1.2 million** [950 000–1.5 million]	**550 000** [430 000–720 000]
Women (15+ years)	21.0 million [19.0–23.5 million]	530 000 [410 000–710 000]	240 000 [180 000–320 000]
Men (15+ years)	18.5 million [16.5–20.7 million]	650 000 [530 000–830 000]	320 000 [240 000–410 000]
Children (<15 years)	**1.4 million** [1.1–1.8 million]	**120 000** [82 000–170 000]	**75 000** [50 000–110 000]

Source: UNAIDS/WHO estimates, 2025.

Estimated HIV infections in the US by region, 2022*

Nearly half (49%) of new HIV infections were in the South.



* Among people aged 13 and older.

Source: CDC. Estimated HIV incidence and prevalence in the United States, 2018–2022. *HIV Surveillance Supplemental Report*, 2024; 29(1).

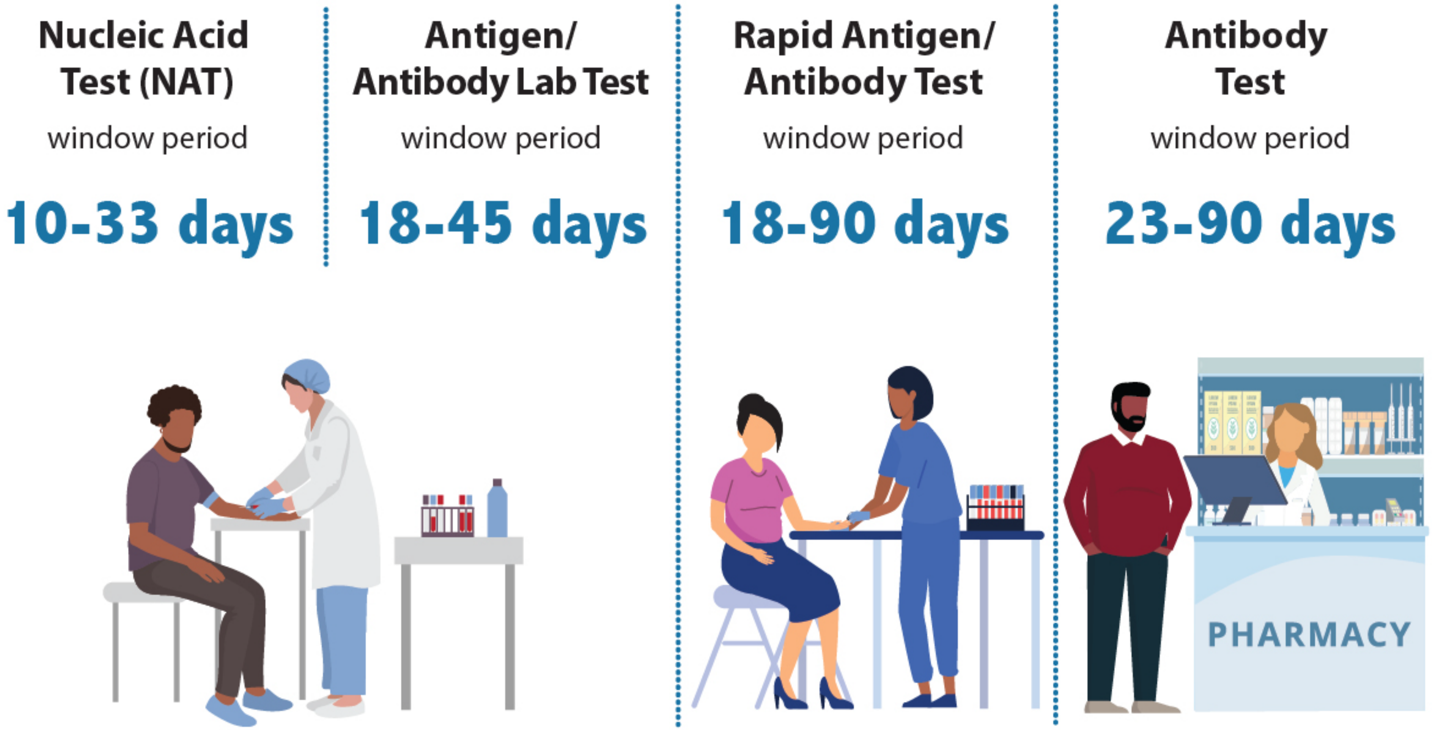
Human Immunodeficiency Virus (HIV)

Testing

- Acute
- Chronic
- AIDS

Test positive for the virus

CD4 count of fewer than 200 cells per microliter of blood (normal 500-1500)



Human Immunodeficiency Virus (HIV)

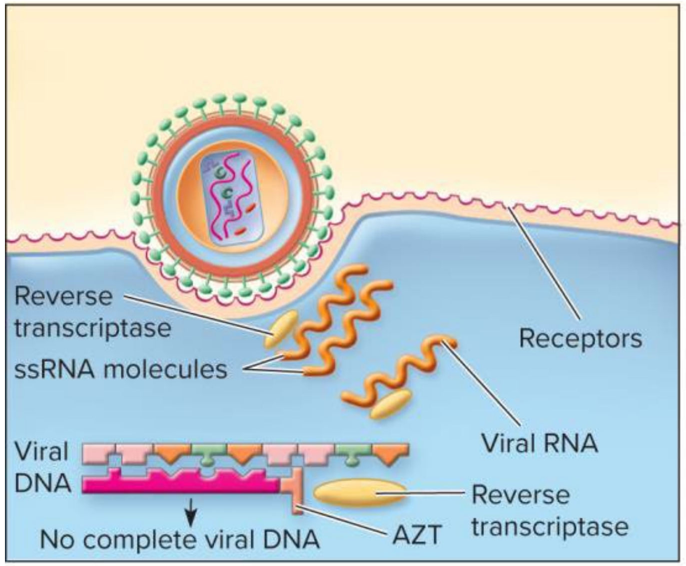
- Prevention
- Barrier protection should be used when having sex with anyone whose HIV status is unknown
- Never sharing needles, syringes, or other drug injection equipment (even if cleaned with bleach)
- Using PrEP (pre-exposure prophylaxis) and PEP (post-exposure prophylaxis).

Human Immunodeficiency Virus (HIV)

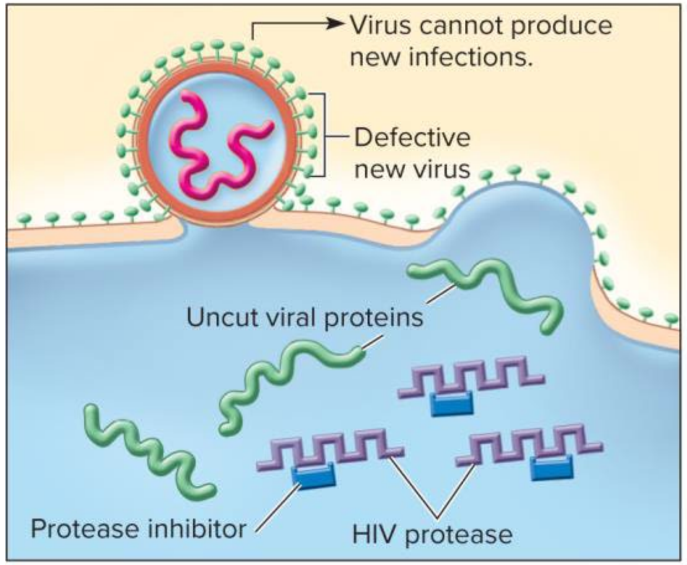
- Treatment

Location of reaction

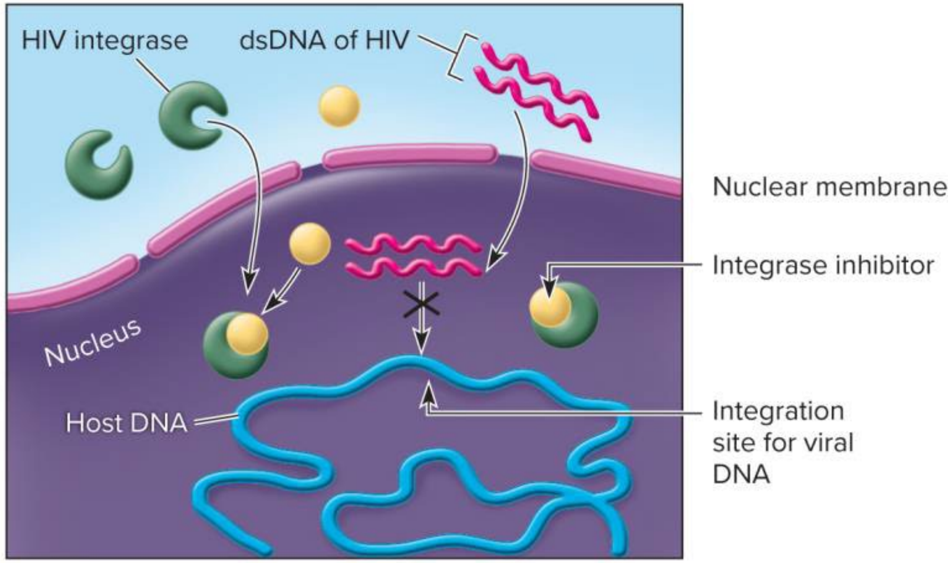
External to cell
Cytoplasm
Nucleus



(a) A prominent group of drugs (AZT, ddI, 3TC) are **nucleoside analogs that inhibit reverse transcriptase**. They are inserted in place of the natural nucleotide by reverse transcriptase but block further action of the enzyme and synthesis of viral DNA. Non-nucleoside RT inhibitors are also in use.



(b) **Protease inhibitors** plug into the active sites on HIV protease. This enzyme is necessary to cut elongated HIV protein strands and produce functioning smaller protein units. Because the enzyme is blocked, the proteins remain uncut, and abnormal defective viruses are formed.



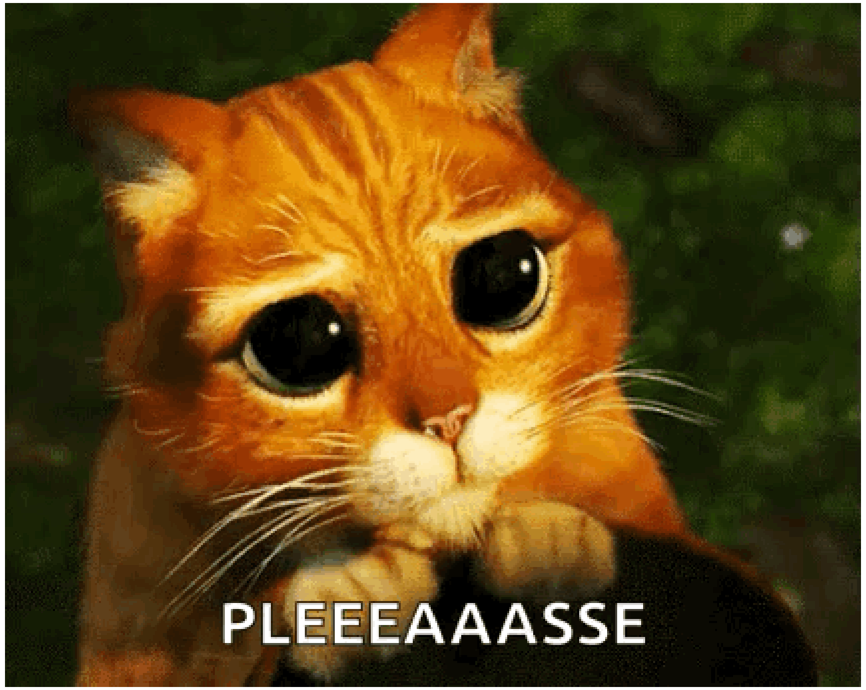
(c) **Integrase inhibitors** are a class of experimental drugs that attach to the enzyme required to splice the dsDNA from HIV into the host genome. This will prevent formation of the provirus and block future virus multiplication in that cell.

HIV disease Table

Causative Organism(s)	Human immunodeficiency virus 1 or 2
Most Common Modes of Transmission	Direct contact (sexual), parenteral (blood-borne), vertical (perinatal and via breast milk)
Virulence Factors	Attachment, syncytia formation, reverse transcriptase, high mutation rate
Culture/Diagnosis	Immunoassay to detect antibodies as well as HIV antigen
Prevention	Avoidance of contact with infected sex partner, contaminated blood, breast milk; pre-exposure prophylaxis (PreP) for high-risk individuals
Treatment	Antiretroviral regimen begun as early as possible
Epidemiological Features	United States: HIV infection = 1.2 million; internationally, HIV infection = 38 million

Thank you

Please take a moment to fill my evaluation form =)



Questions? Suggestions? Opinions?
cilicia.nascimento@bcm.edu